Predicting gene function from gene expression trends, protein features and cis-regulatory information – A rough set modeling approach

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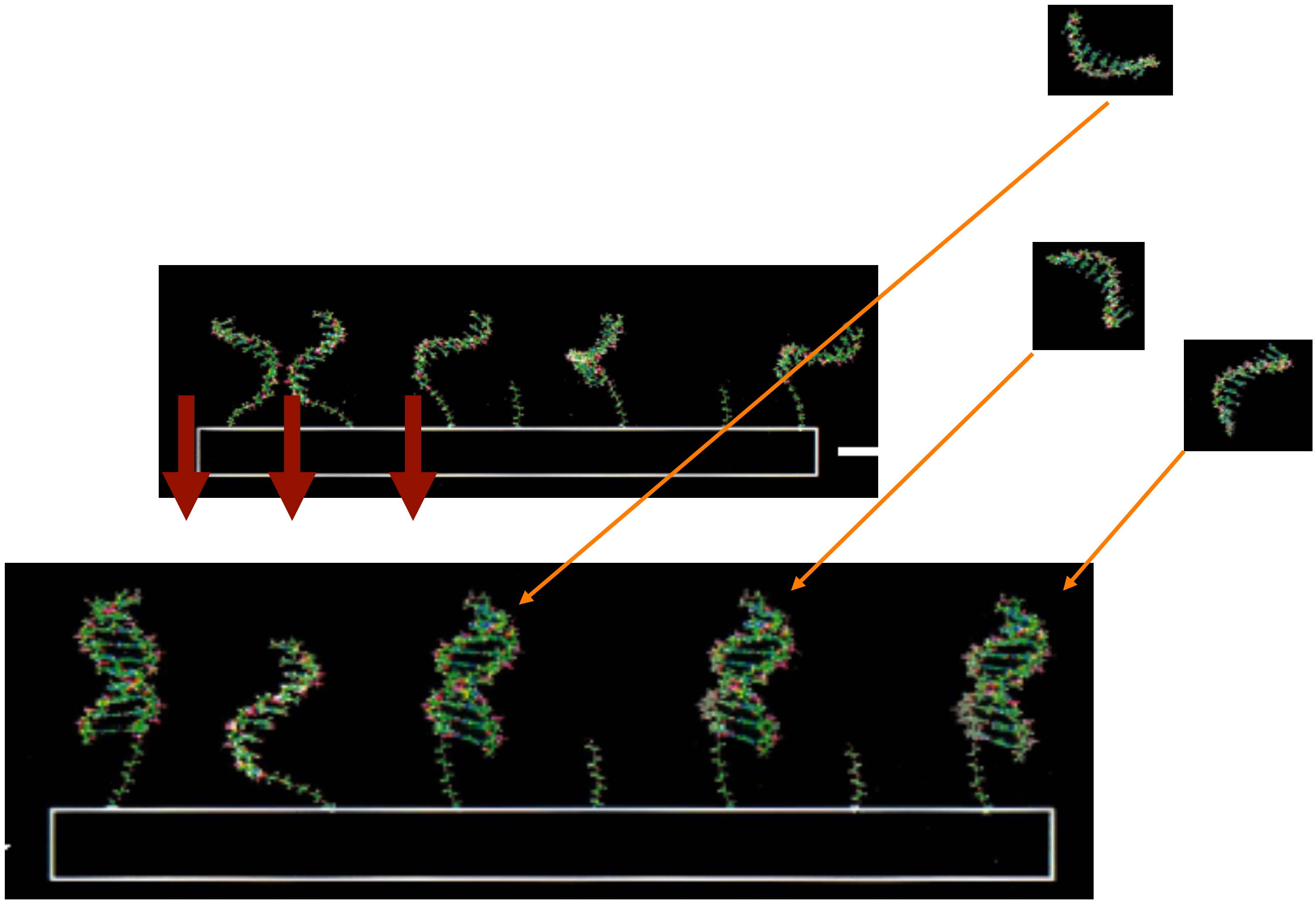
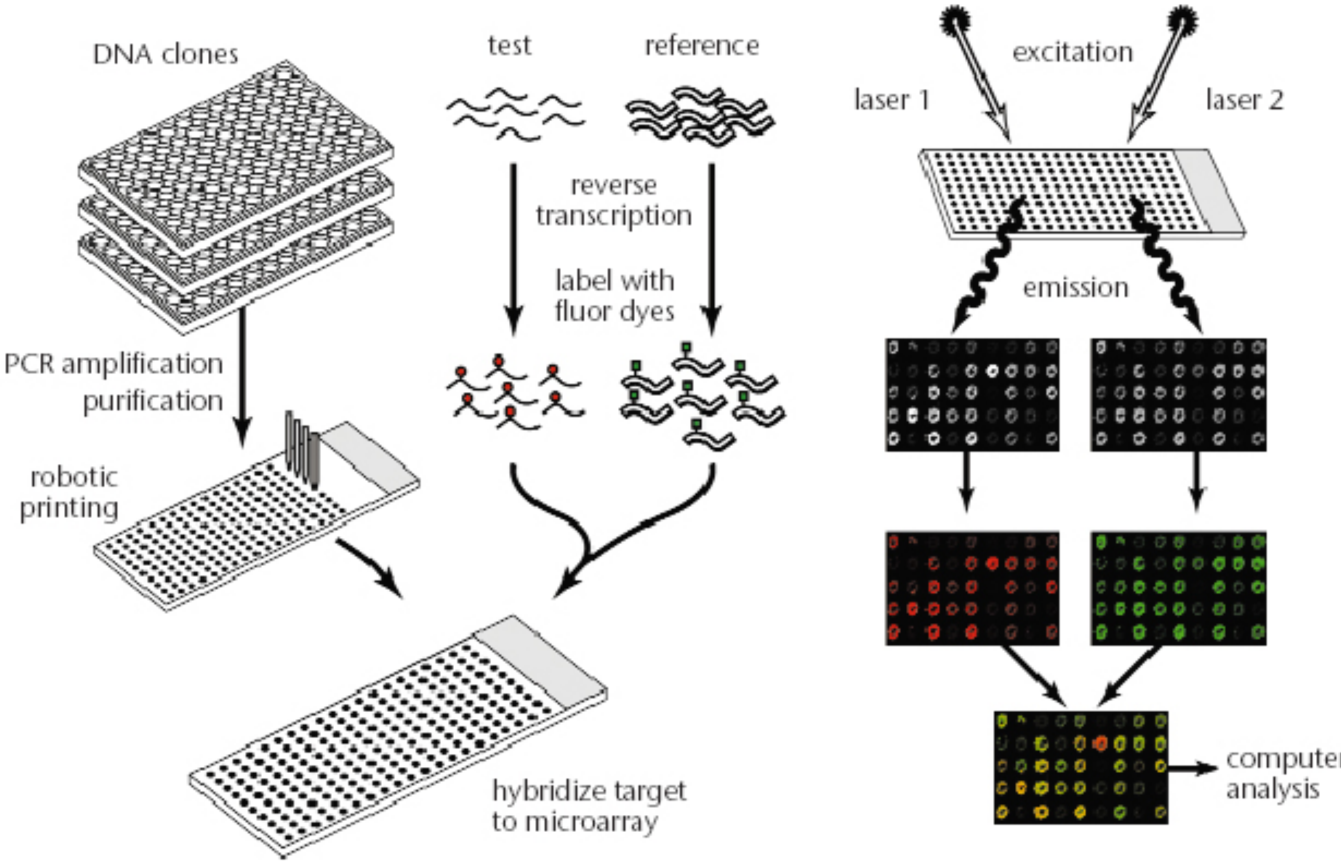
Hybridization



Microarray



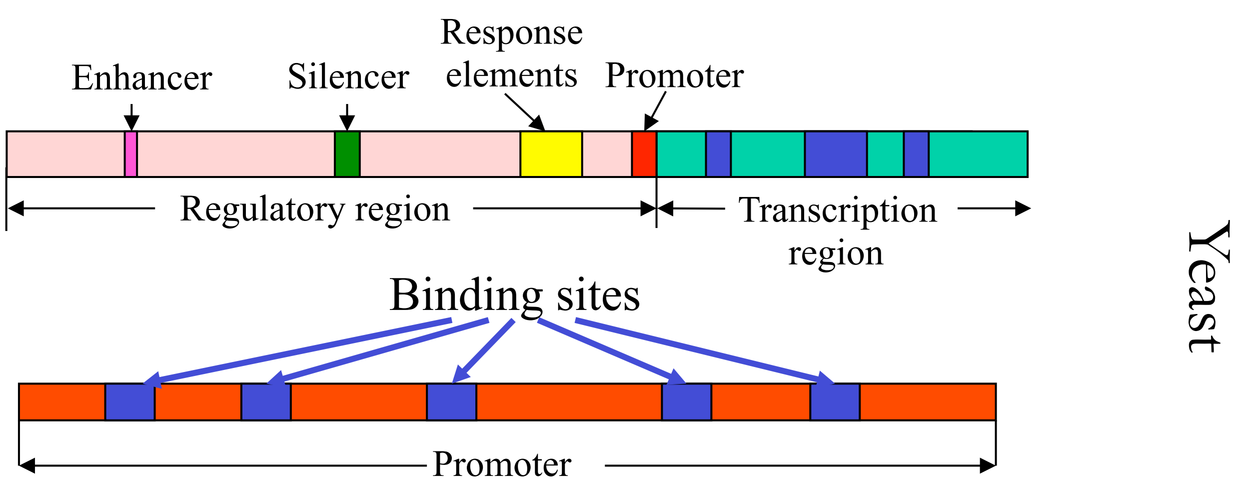
DNA clones
PCR amplification
purification
robotic
printing
test
reference
reverse
transcription
label with
fluor dyes
hybridize target
to microarray
excitation
laser 1
laser 2
emission
computer
analysis

Regulatory logics

- Massive readouts of cell content in terms of RNA molecules (transcriptomics), proteins (proteomics) and the products of metabolic processes (metabolomics) can be explained by the regulatory logics hard-wired in the DNA sequence
- Regulation is organized in modules of genes often participating in the same biological process

Gene regulation

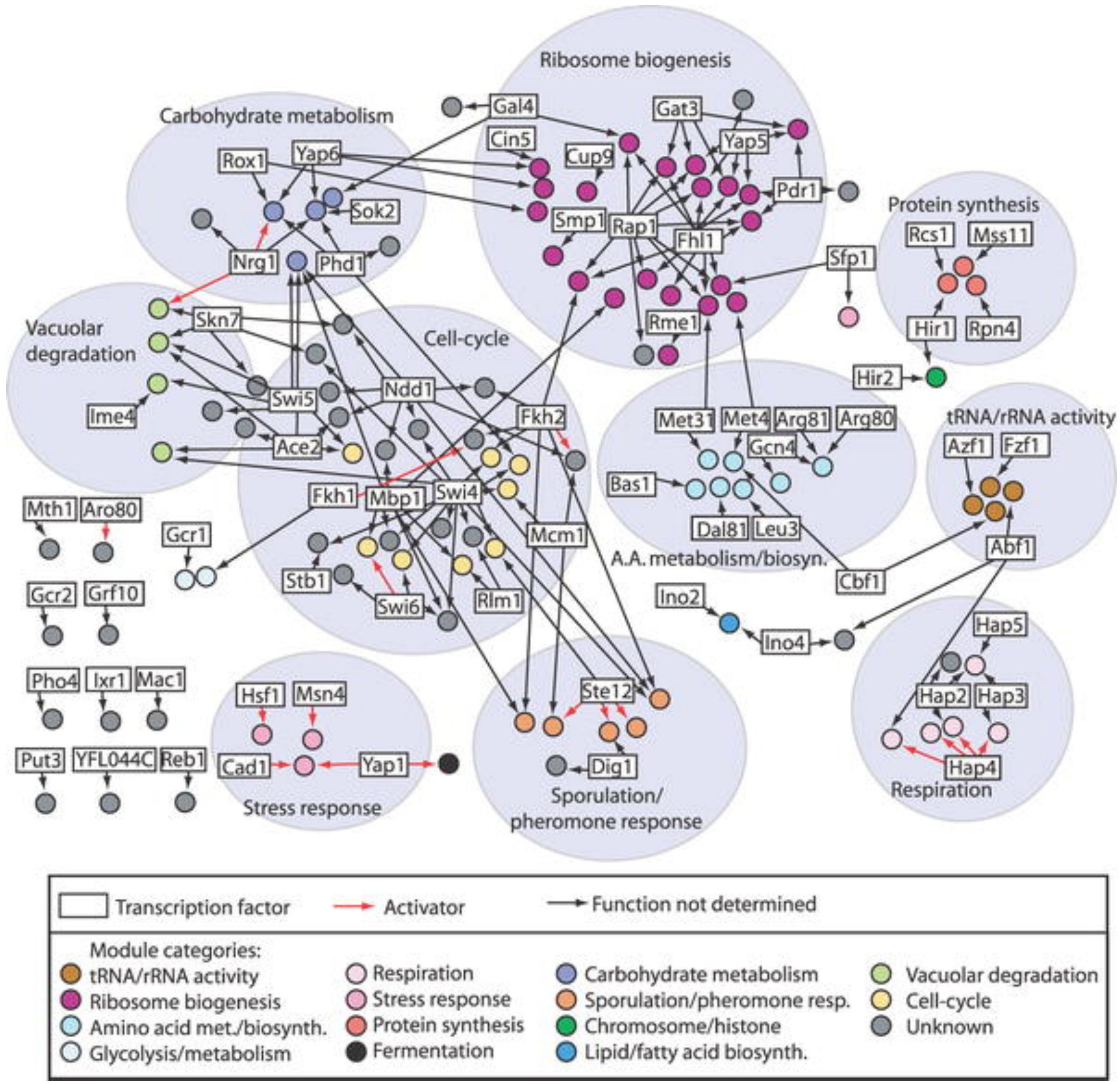
- Gene expression is regulated by regulatory proteins (*transcription factors*)
- Transcription factors depend on recognizing sequence motifs (*binding sites*) in order to effect the expression of genes
- Transcription factors combine to respond to a large number of stress factors (e.g. heat shock) with a large number of expression outcomes



Regulatory logics

- Massive readouts of cell content in terms of RNA molecules (transcriptomics), proteins (proteomics) and the products of metabolic processes (metabolomics) can be explained by the regulatory logics hard-wired in the DNA sequence
- Regulation is organized in modules of genes often participating in the same biological process

Yeast regulatory modules

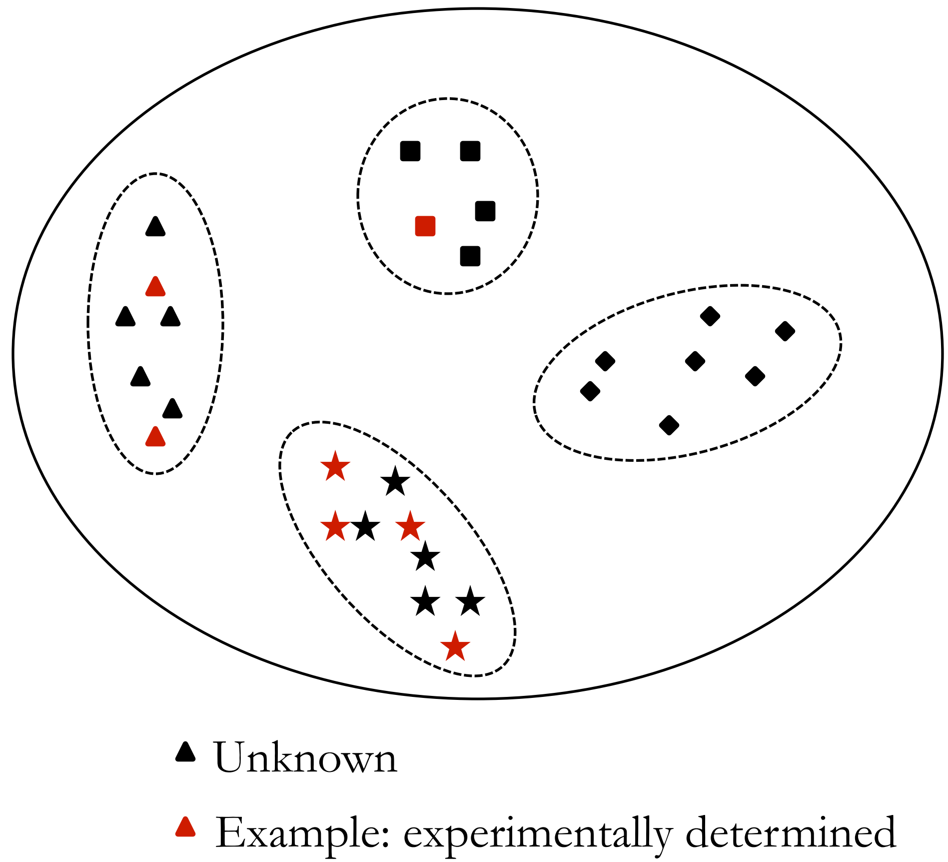


From: Modularity and Dynamics of Cellular Networks Qi Y, Ge H PLoS Computational Biology Vol. 2, No. 12, e174, 2006.

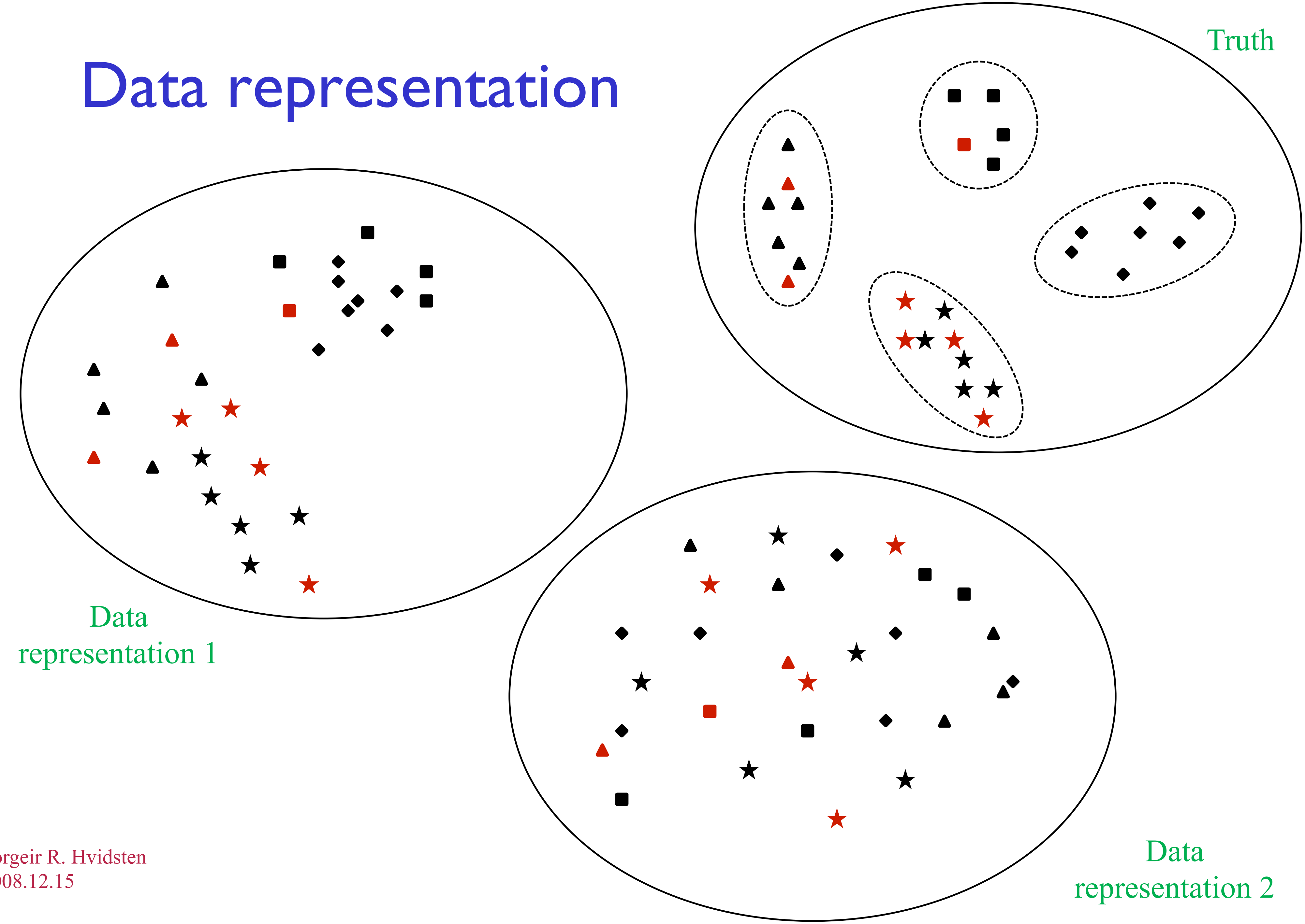
The machine learning strategy …

… iteratively uses experiments to provide representative examples and computational models to provide experimentalists with new, testable hypotheses

- Nearest neighbor predictors
  - evolutionary link
  - need few examples
- Model inducers
  - more powerful
  - interpretable models



Data representation



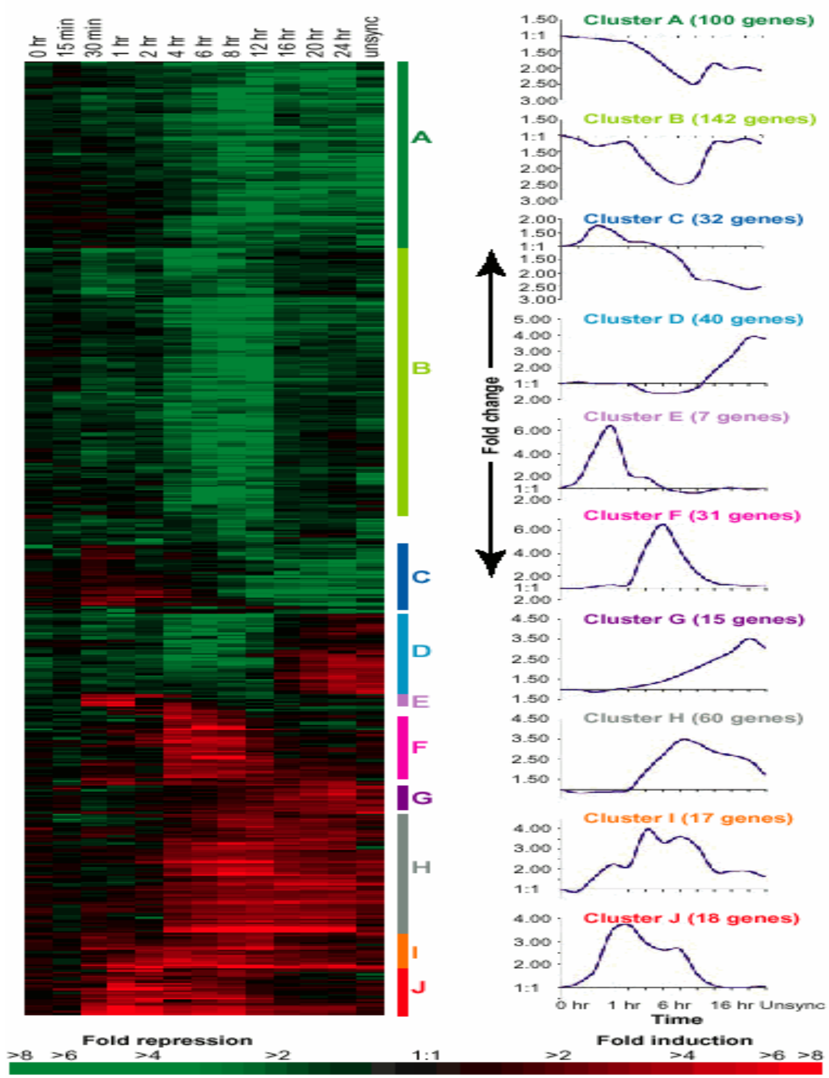
Predicting biological process from gene expression time profiles

Papers:

I. T. R. Hvidsten, A. Lægreid and J. Komorowski. Learning rule-based models of biological process from gene expression time profiles using gene ontology, *Bioinformatics* 19(9): 1116-23, 2003.

II. A. Lægreid, T. R. Hvidsten, H. Midelfart, J. Komorowski and A. K. Sandvik. Predicting Gene Ontology Biological Process From Temporal Gene Expression Patterns, *Genome Research*, 13(5): 965-979, 2003.

Hierarchical clustering

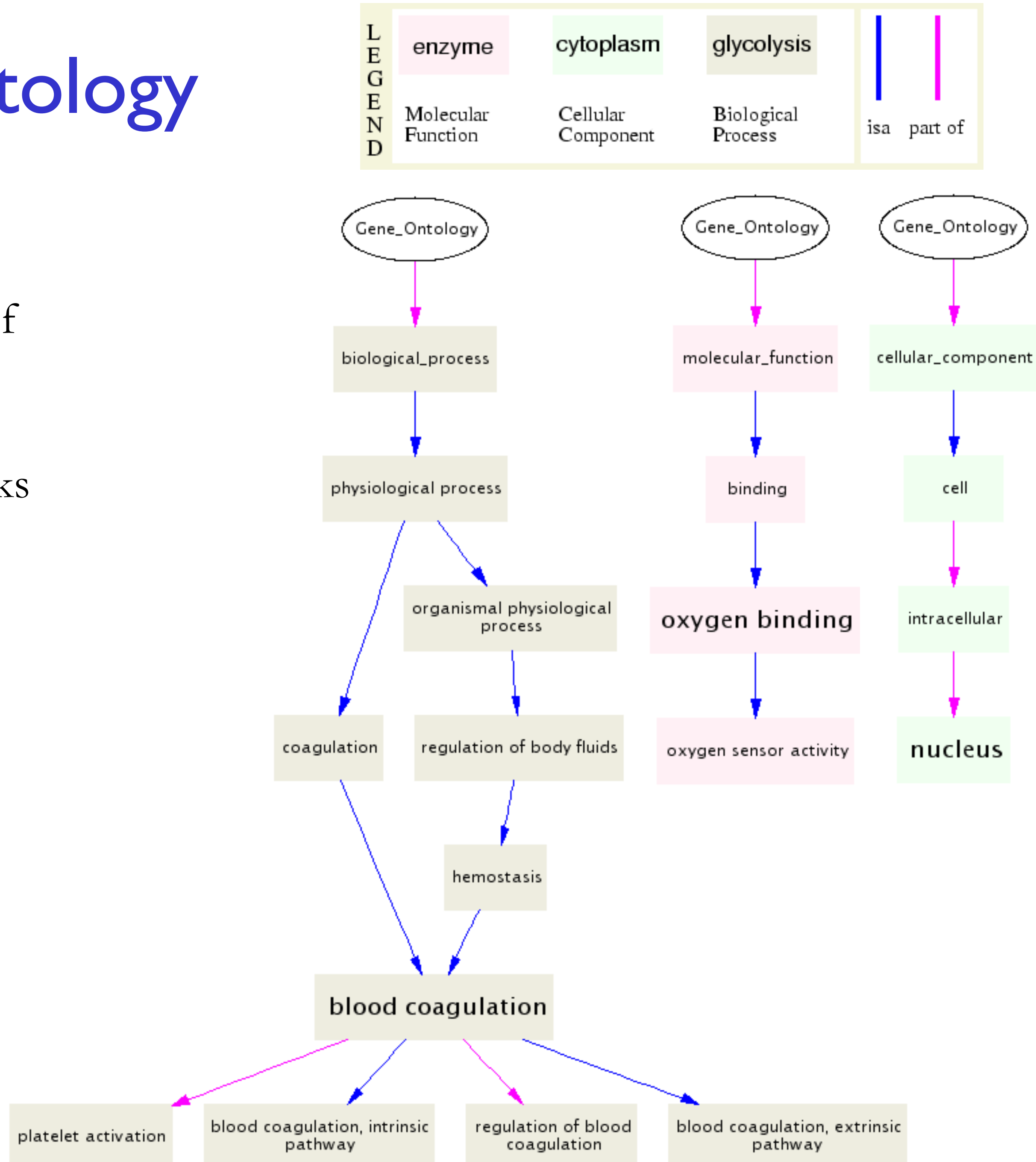


Iyer et al., The transcriptional program in the response of human fibroblasts to serum, *Science*, 283(5398): 83-87, 1999

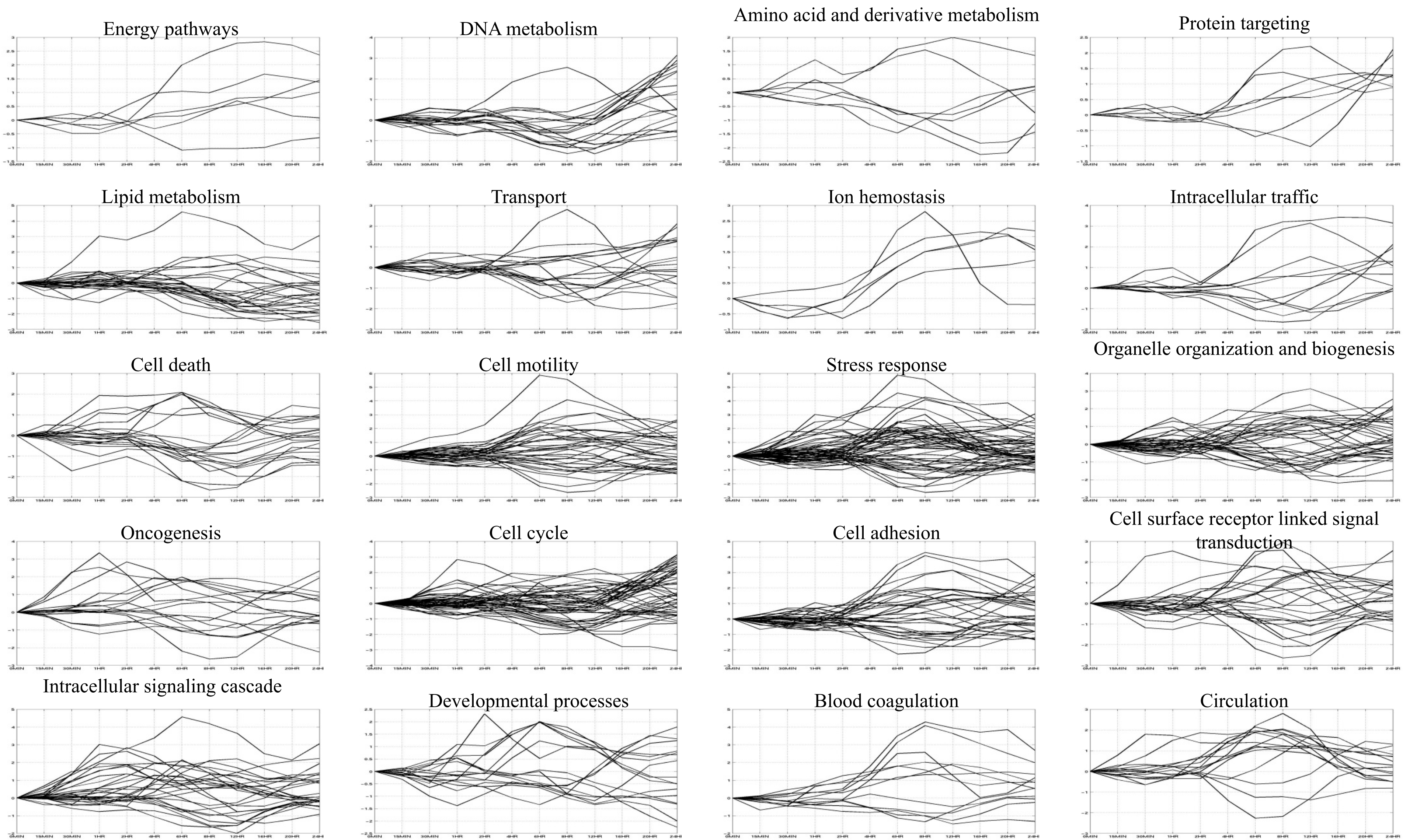
Gene Ontology

Ordered controlled vocabulary organized in a taxonomy for describing the molecular role of gene products

- Molecular function: the tasks performed by individual gene products
- Biological process: broad biological goals that are accomplished by ordered assemblies of molecular functions
- Cellular component: subcellular structures, locations, and macromolecular complexes



Biological processes from GO



Torg
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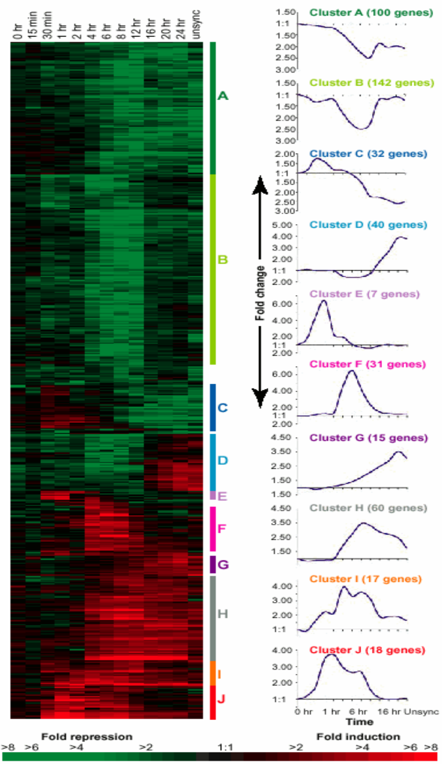
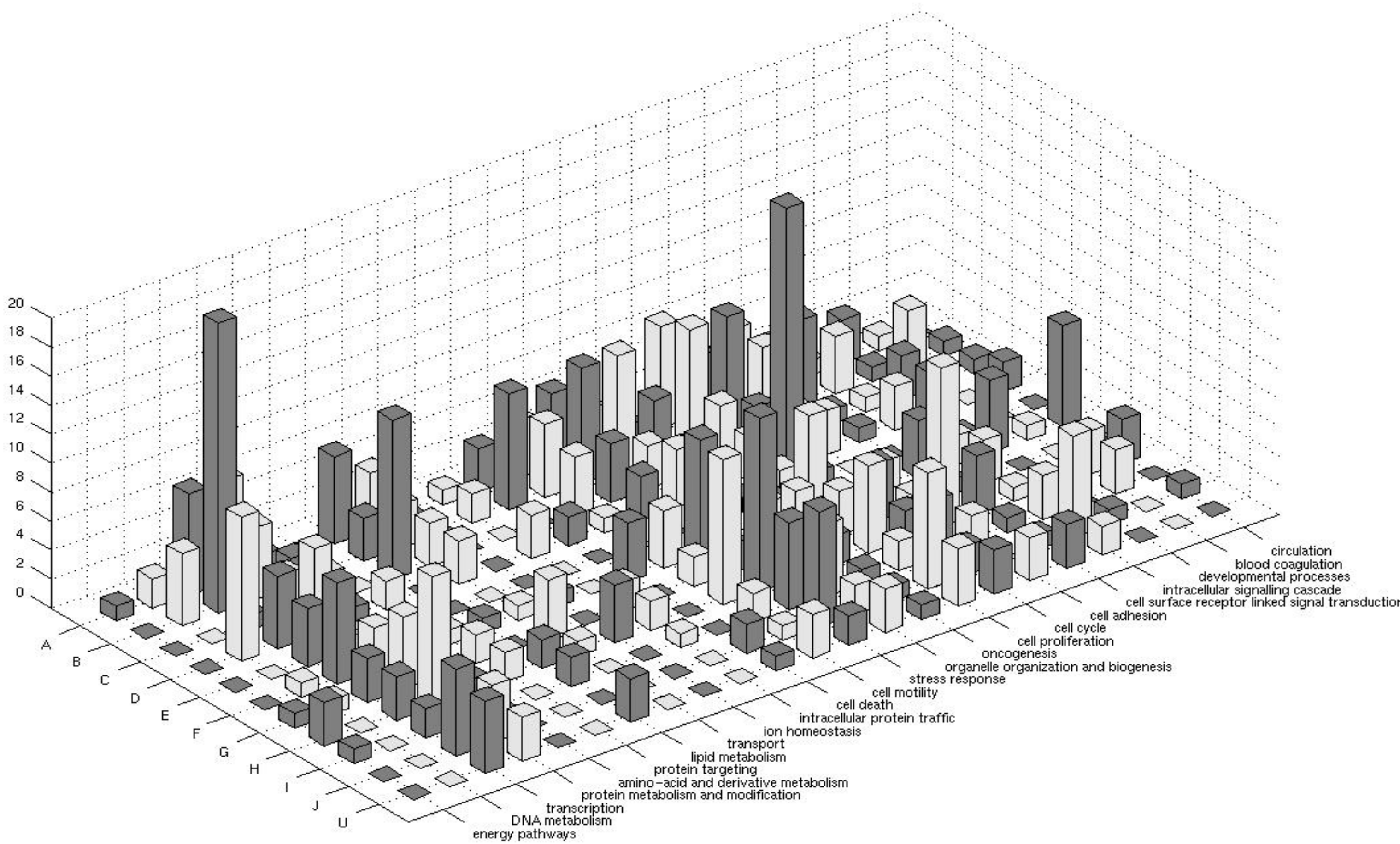
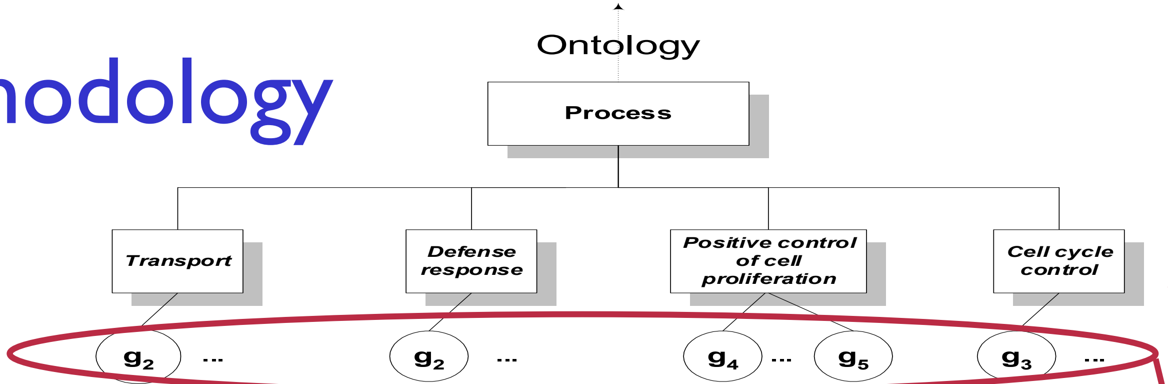
Gene Ontology vs. expression clustering





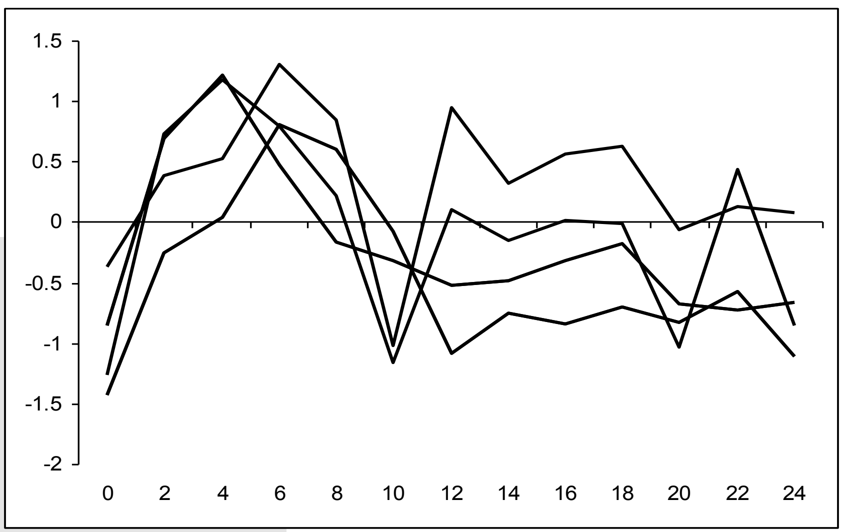
Methodology



1. Annotation

Gene	0HR	15MIN	30MIN	1HR	2HR	4HR	6HR	8HR	12HR	16HR	20HR	24HR	Process
g_1	0.00	-0.47	-3.32	-0.81	0.11	-0.60	-1.36	-1.03	-1.84	-1.00	-0.60	-0.94	Unknown
g_2	0.00	0.66	0.07	0.20	0.29	-0.89	-0.45	-0.29	-0.29	-0.15	-0.45	-0.42	Transport and defense response
g_3	0.00	0.14	-0.04	0.00	-0.15	-0.58	-0.30	-0.18	-0.38	-0.49	-0.81	-1.12	Cell cycle control
g_4	0.00	-0.04	0.00	-0.23	-0.25	-0.47	-0.60	-0.56	-1.09	-0.71	-0.76	-0.62	Positive control of cell proliferation
g_5	0.00	0.28	0.37	0.11	-0.17	-0.18	-0.60	-0.23	-0.58	-0.79	-0.29	-0.74	Positive control of cell proliferation
...	...	...	...	...	...	...	...	...	...	...	...	...	...

2. Extracting features for learning



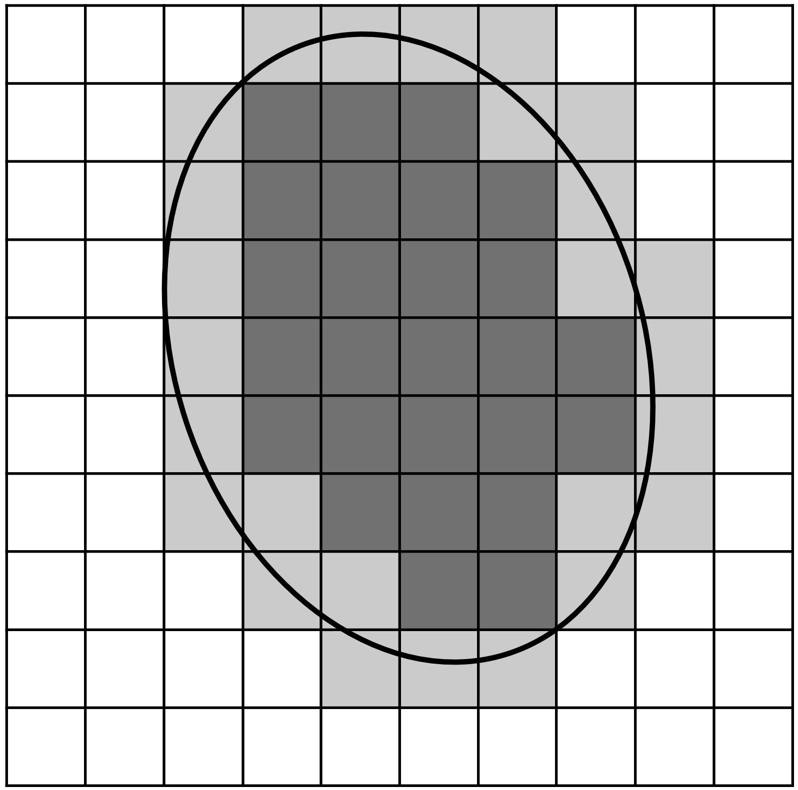
3. Inducing minimal decision rules using rough sets

0 - 4(Increasing) AND 6 - 10(Decreasing) AND 14 - 18(Constant) => GO(cell proliferation)

4. The function of uncharacterized genes is predicted using the rules

!

Rough set



- A mathematical theory for viewing data in terms of sets of indiscernible objects (equivalence classes)
- A rough set X
  - lower approximation: $\underline{A}X$
  - upper approximation: $\overline{A}X$
  - $\overline{A}X - \underline{A}X \neq \emptyset$
- A crisp set
  - $\overline{A}X - \underline{A}X = \emptyset$

Rule Induction

IF 0 - 4(Constant) AND 0 - 10(Increasing)

THEN GO(prot. met. and mod.) OR GO(mesoderm develop.)
OR GO(prot. biosynt.)

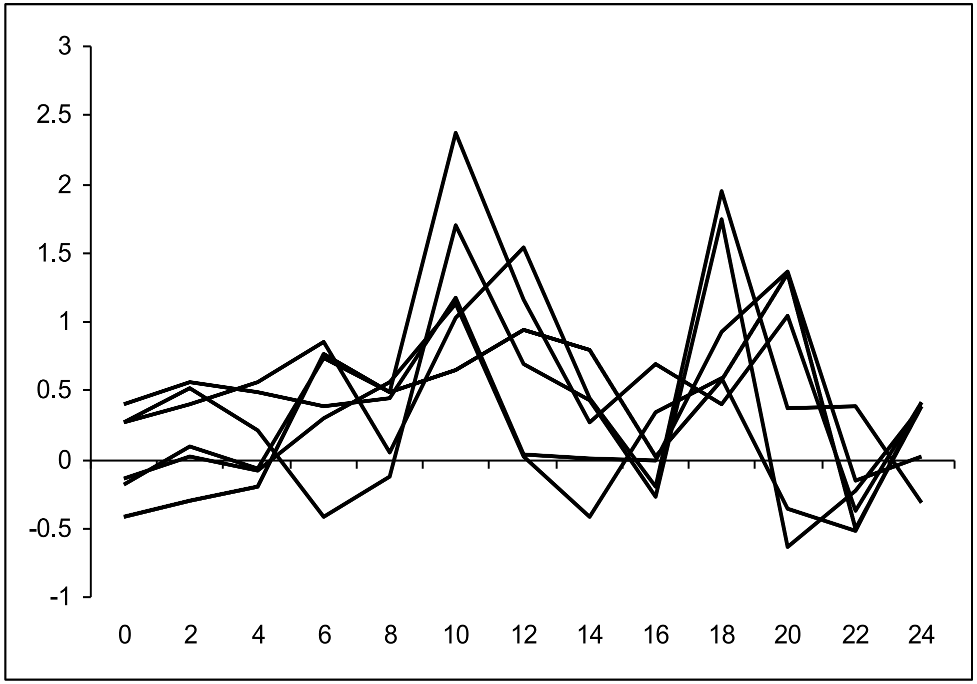
- IF-part (antecedent, premise): the minimal set of discrete changes in expression needed to uphold the discriminatory power of the full data set
- THEN-part (consequent): all functions of genes described by the premise-side
- We want rules that describe the expression profiles of several genes with one or a few functions
  - accuracy: the fraction of genes matching the IF-part that are annotated with the process in the THEN-part
  - coverage: the fraction of genes annotated with the process in the THEN-part that matches the IF-part

Genetic algorithm for reduct computation

- Individuals in the population are subsets of time intervals: 00110 … *(absent, absent, present, present, absent, …)*
- Fitness:
  - A: Fraction of genes with different function that can be discerned from the function of interest

  +

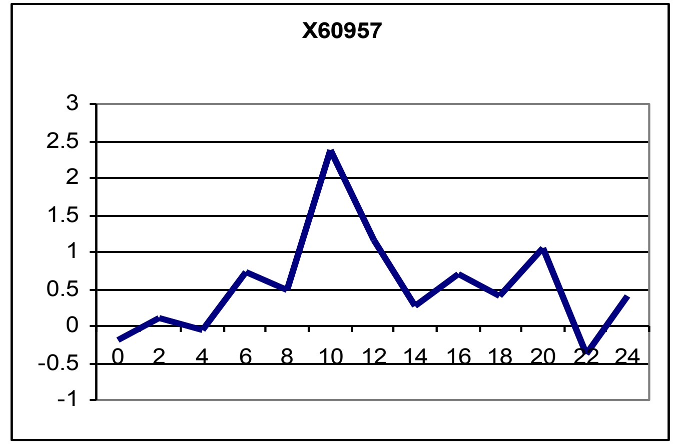
  - B: Number of time intervals that are *absent*
- Typically, approximate solutions are obtained, i.e. A < 1.0



Rule example

Rule	**Covered genes**
0 - 4(Constant) AND 0 - 10(Increasing) =>	
GO(protein metabolism and modification) OR	**M35296 J02783 D13748 X05130**
GO(mesoderm development) OR	X60957
GO(protein biosynthesis)	D13748

Classification



IF … THEN …
IF … THEN …
IF … THEN …
IF … THEN …
IF … THEN …
IF … THEN …
IF … THEN …

IF 0 - 4(Constant) AND 0 - 10(Increasing) THEN
GO(protein metabolism and modification) OR
GO(mesoderm development) OR
GO(protein biosynthesis)

+4
+1
+1

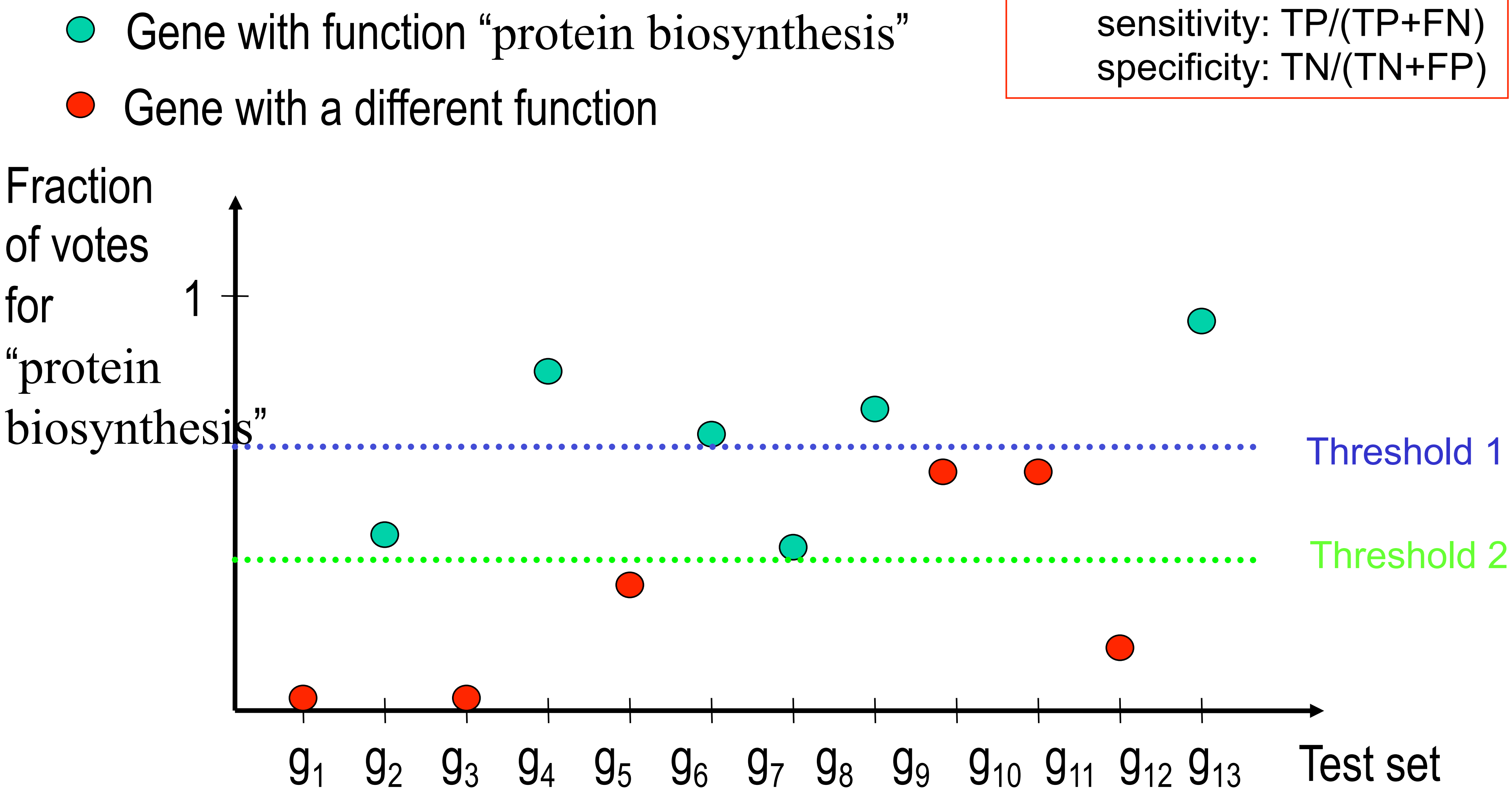
IF … THEN
IF … THEN …
IF … THEN …
IF … THEN …
IF … THEN …

Process	Votes
protein metabolism and modification	6
mesoderm development	3
proteolysis and peptidolysis	2
transcription	1
protein biosynthesis	1
vision	1
…	

Votes are normalized and processes with vote fractions higher than a selection-threshold are chosen as predictions

Threshold selection

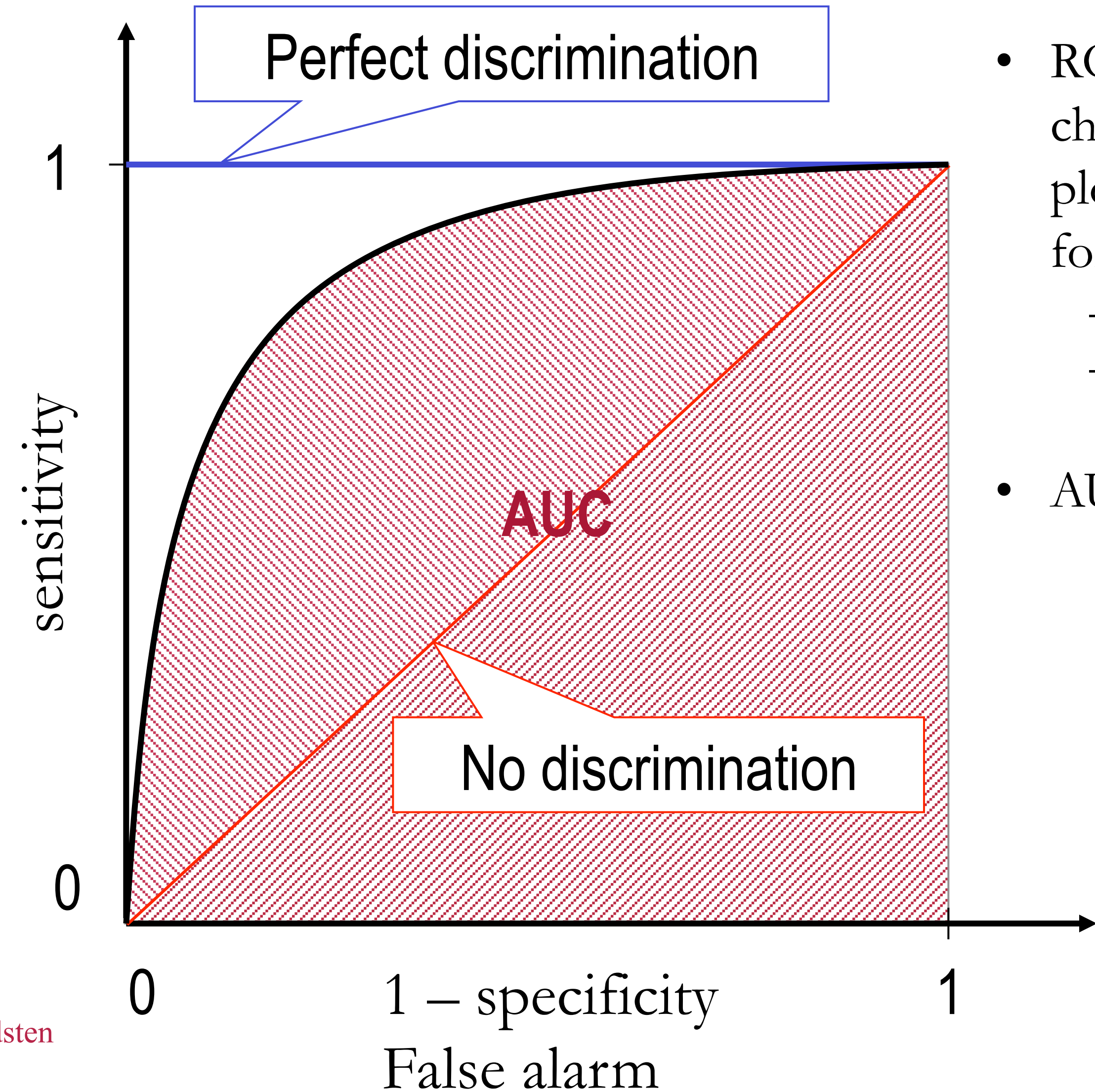
sensitivity: TP/(TP+FN)
specificity: TN/(TN+FP)



Sensitivity = 2/3, Specificity=1

Sensitivity = 1, Specificity=2/3

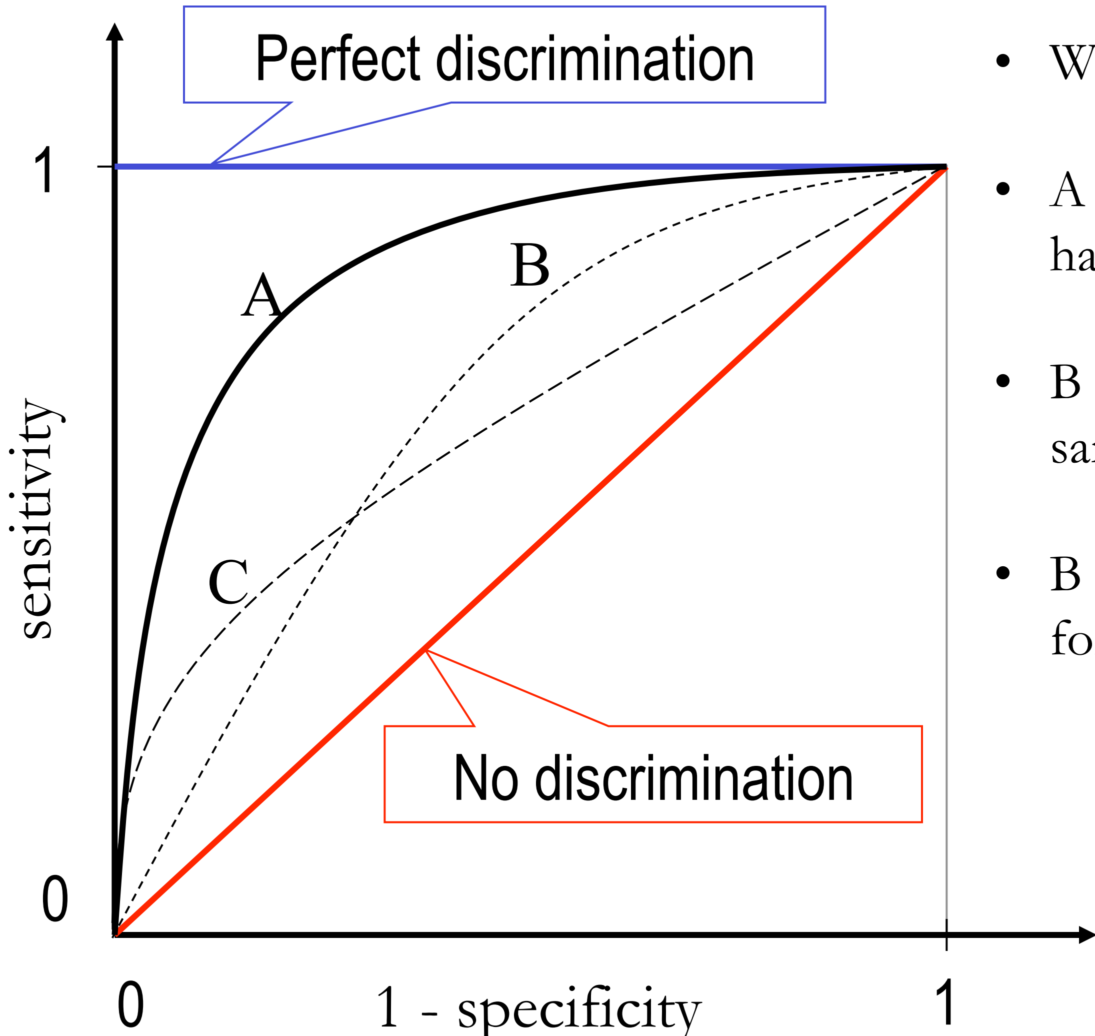
ROC analysis and classifier evaluation



- ROC: Receiver operating characteristics curve results from plotting sensitivity against specificity for all possible thresholds
  - sensitivity: TP/(TP+FN)
  - specificity: TN/(TN+FP)

- AUC: Area under the ROC curve

ROC analysis and classifier evaluation



- Which ROC curve is better?
- A dominants B and C and clearly has a higher AUC
- B and C have approximately the same AUC
- B is better for some thresholds, C for others

Cross validation estimates*

Treshold independent:

PROCESS	AUC	SE	P-VALUE
Ion homeostasis	1.00	0.00	0.008
Protein targeting	0.99	0.03	0.000
Blood coagulation	0.96	0.08	0.000
DNA metabolism	0.94	0.09	0.000
Intracellular signaling cascade	0.94	0.06	0.000
Cell cycle	0.93	0.04	0.000
Energy pathways	0.93	0.12	0.004
Oncogenesis	0.92	0.11	0.000
Circulation	0.91	0.11	0.001
Cell death	0.90	0.10	0.000
Developmental processes	0.90	0.07	0.000
Defense (immune) response	0.88	0.05	0.000
Transcription	0.88	0.11	0.002
Cell adhesion	0.87	0.09	0.002
Stress response	0.86	0.15	0.002
Protein metabolism and modification	0.85	0.10	0.000
Cell motility	0.84	0.11	0.000
Cell surface rec linked signal transd	0.82	0.15	0.005
Lipid metabolism	0.81	0.14	0.000
Cell organization and biogenesis	0.79	0.11	0.000
Cell proliferation	0.79	0.06	0.002
Transport	0.79	0.17	0.001
Amino acid and derivative metabolism	0.69	0.06	0.288
AVERAGE	**0.88**	**0.09**	

Selecting a selection threshold:

Over all classes:

Coverage/recall = TP/(TP+FN)

Precision = TP/(TP+FP)

Coverage: 84%
Precision: 50%

Coverage: 71%
Precision: 60%

Coverage: 39%
Precision: 90%

* V. R. Iyer et al. The transcriptional program in the response of human fibroblasts to serum, *Science*, 283(5398): 83-87, 1999.

Reclassifying annotated genes

12 of 24 ”false positive” predictions for oncogenesis was ”missing annotations” found through litterature search

Symbol	Gene name	Molecular function	Comment	Reference (PMID)
CCNG1	Cyclin G1	CDK kinase regulator	p53 target	11327114
CDKN1C	cyclin-dependent kinase inhibitor 1C	cyclin-dependent protein kinase inhibitor	tumor suppressor	7729684
CAT	catalase	oxidoreductase	tumor progression	8513880,
ALDH3A2	aldehyde dehydrogenase 10	aldehyde dehydrogenase	tumor progression	92393980
ADD3	adducin 3 (gamma)	membrane-cytoskeleton-associated protein	tumor progression	9607561
TFDP2	transcription factor Dp-2 (E2F dimerization partner 2)	transcription co-factor	cell cycle regulation	7784053
ATRX	alpha thalassemia/mental retardation syndrome	DNA helicase	transcription & DNA repair	10362365
EPS15	epidermal growth factor receptor pathway substrate 15	kinase substrate	growth regulation	93361014
EGR1	early growth response 1	transcription factor	tumor suppressor	9109500
NR4A2	nuclear receptor subfam 4, group A, m2 (Nurr1, Not)	ligand-dependent nuclear receptor	proto-oncogene	9592180
NR4A3	nuclear receptor subfam 4, group A, m 3 (Nor1)	ligand-dependent nuclear receptor	proto-oncogene	9592180
COPEB	core promoter element binding protein	transcription factor	proto-oncogene	9268646

Prediction of uncharacterized genes: 2008

- 96 (of 211) ”uncharacterized genes” in 2003 has some relevant annotation in 2008
- ~1/3 have correct predictions by our method
- Notes:
  - We predict 23 biological processes (Random guessing = 1/23)
  - Not trivial to map old annotations to new GO tree
  - Also noted in the original Science paper: there is a huge overrepresentation of down-regulated genes among 2003-uncharacterized genes: not a representative training set
- Example: Gene MRE11A correctly predicted to *DNA metabolic process*, *cell cycle* and *cellular component organization*

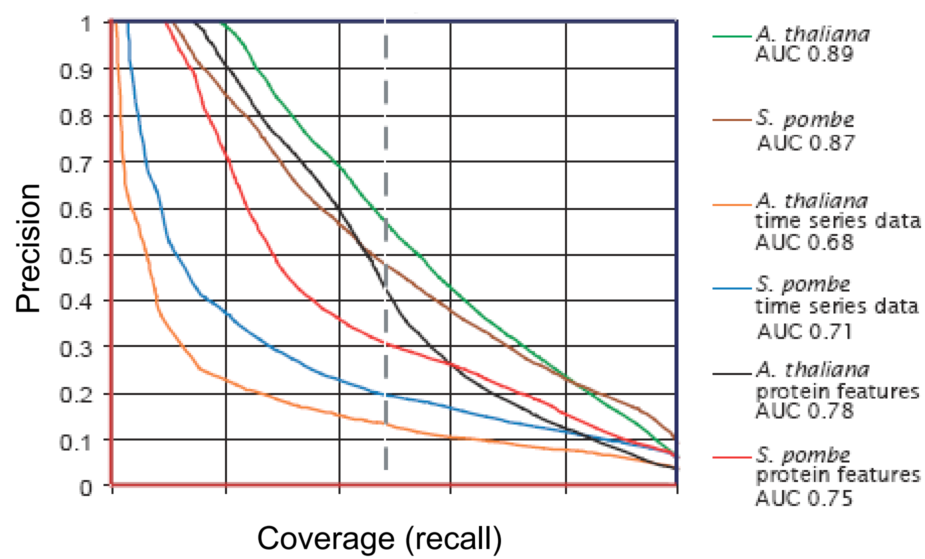
Predicting biological process from gene expression time profiles – including protein features

Papers:

I. K. Wabnik, T. R. Hvidsten, A. Kedzierska, J. Van Leene, G. De Jaeger, G. T. S. Beemster, J. Komorowski and M. T. R. Kuiper. Gene expression trends and protein features effectively complement each other in gene function prediction, Accepted to Bioinformatics, 2008.

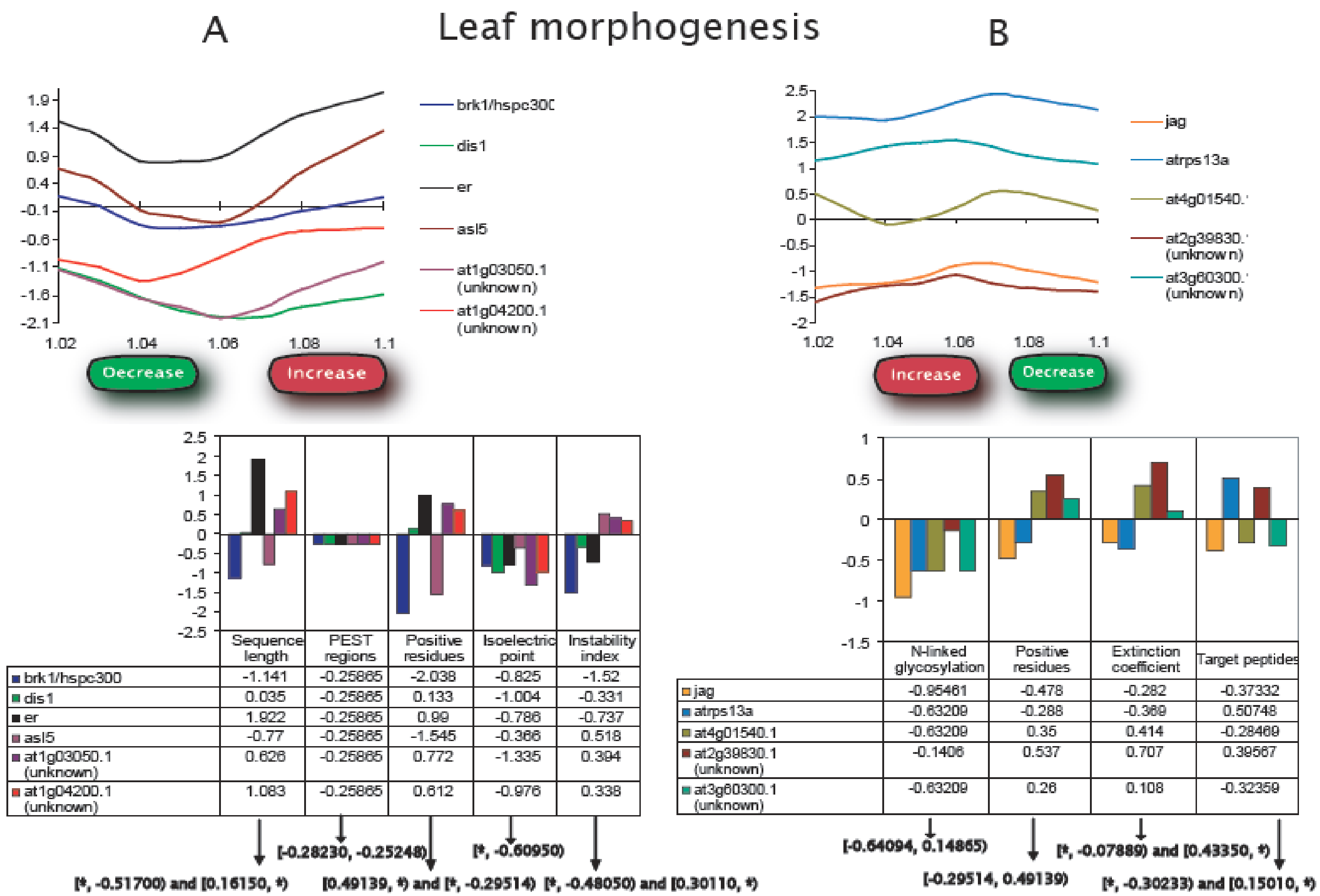
Results

- Protein features: derived directly from amino acid*
- Synergy from combining expression and protein sequence features
- Expression data increases preformance for relevant biological processes
- Extention: Pick the most significant/ certain predictions from each dataset from many diverse sets



*U. Lichtenberg, T. Jensen , L. Jensen and S. Brunak . Protein feature based identification of cell cycle regulated proteins in yeast. *J Mol Biol*, **329**, 149-170, 2003.

Rule interpretation



	Sequence length	PEST regions	Positive residues	Isoelectric point	Instability index
brk1/hspc300	-1.141	-0.25865	-2.038	-0.825	-1.52
dis1	0.035	-0.25865	0.133	-1.004	-0.331
er	1.922	-0.25865	0.99	-0.786	-0.737
asl5	-0.77	-0.25865	-1.545	-0.366	0.518
at1g03050.1 (unknown)	0.626	-0.25865	0.772	-1.335	0.394
at1g04200.1 (unknown)	1.083	-0.25865	0.612	-0.976	0.338

	N-linked glycosylation	Positive residues	Extinction coefficient	Target peptides
jag	-0.95461	-0.478	-0.282	-0.37332
atrps13a	-0.63209	-0.288	-0.369	0.50748
at4g01540.1	-0.63209	0.35	0.414	-0.28469
at2g39830.1 (unknown)	-0.1406	0.537	0.707	0.39567
at3g60300.1 (unknown)	-0.63209	0.26	0.108	-0.32359

E.g.: IF 1.03-1.05(decrease) **AND** 1.07-1.1(increase) **AND** sequence length ([*,-0.52) -[0.16,*] **THEN** *leaf morphogenesis*

Discovering regulatory logics

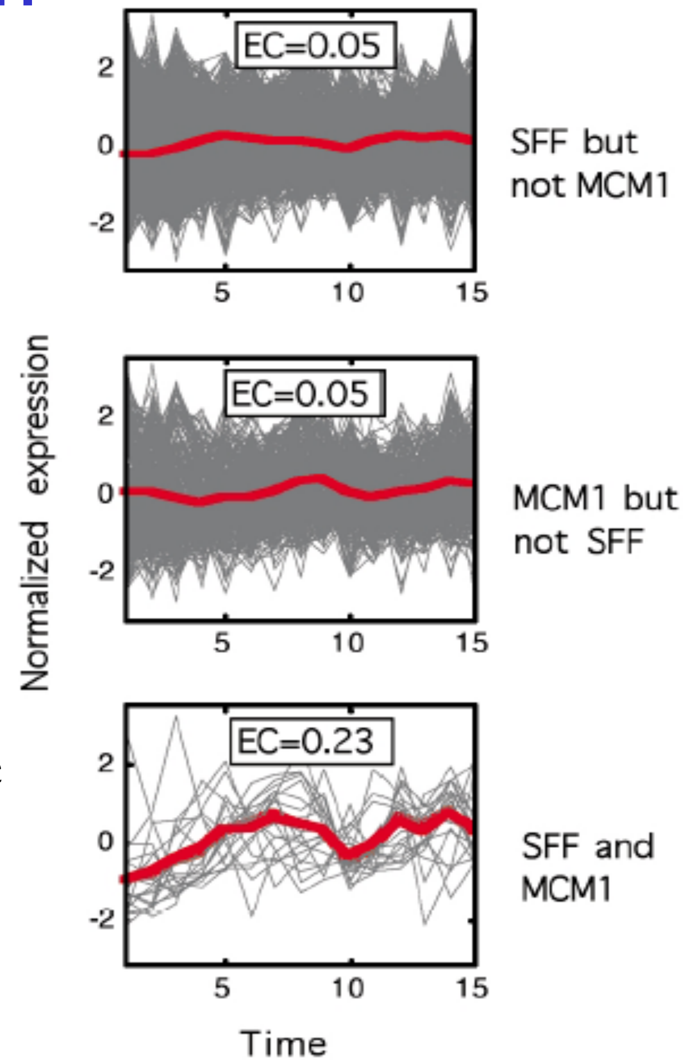
- T. R. Hvidsten, B. Wilczynski, A. Kryshtafovych, J. Tiuryn, J. Komorowski and K. Fidelis. Discovering regulatory binding site modules using rule-based learning. Genome Research 15: 856-66, 2005.
- B. Wilczynski, T. R. Hvidsten, A. Kryshtafovych, J. Tiuryn, J. Komorowski, K. Fidelis. Using Local Gene Expression Similarities to Discover Regulatory Binding Site Modules, BMC Bioinformatics 7: 505, 2006.

Combinatorial regulation

Many studies have used gene expression data to search for overrepresented sequence motifs in co-expressed genes

Pilpel et al. (2001) found that genes sharing pairs of binding sites are significantly more likely to be co-expressed than genes with only single binding sites in common

Expression coherence score (EC)



Pilpel, Y., P. Sudarsanam, and G.M. Church. 2001. Identifying regulatory networks by combinatorial analysis of promoter elements. *Nat Genet* **29:** 153-159.

Rule Induction

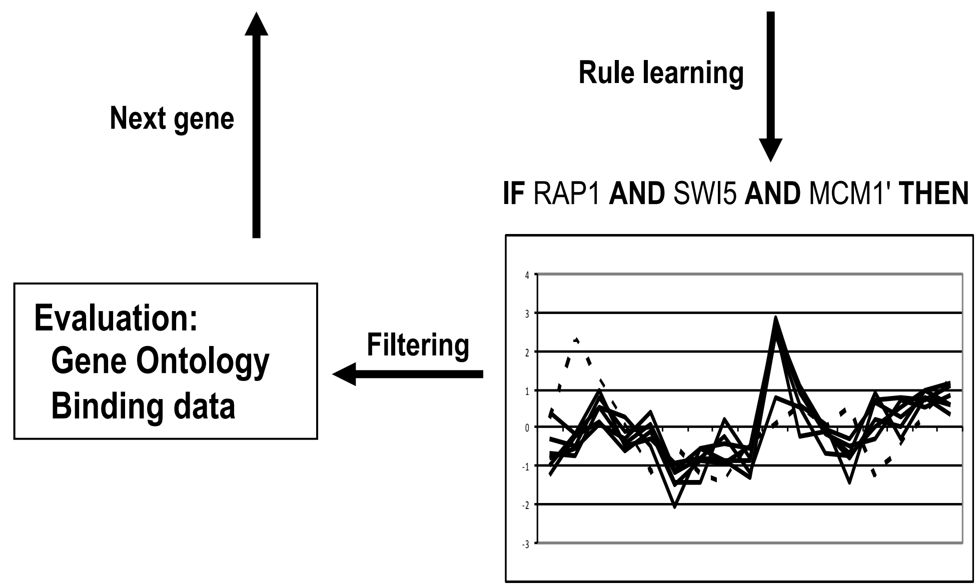
IF binding site i AND binding site j

THEN particular expression profile

- IF-part (antecedent, premise): the minimal set of binding sites needed to uphold the discriminatory power of the full data set
- THEN-part (consequent): an expression profiles
- We want rules that describe a combination of binding sites common to genes with similar expression
  - accuracy: the fraction of genes matching the IF-part that have similar expression to the profile in the THEN-part
  - coverage: the fraction of genes with similar expression to the profile in the THEN-part that matches the IF-part

Data and Method

- a database of 43 known and 313 putative yeast binding site motifs
- expression profiles for yeast genes measured under six different conditions: cell cycle and five stress conditions (sporulation, diauxic shift, heat and cold shock, pheromone and DNA-damaging agents)



Gene	Binding sites						
	HAP234	RAP1	PHO	SWI5	ECB	MCM1'	...
RPL18A	**0**	**1**	**0**	**1**	**0**	**1**	**...**
RPS18A	0	1	0	1	0	1	...
RPL16B	1	1	0	1	0	1	...
RPL26A	0	1	0	1	0	1	...
RPS24A	0	1	0	1	0	1	...
RPL30	0	1	0	1	0	1	...
RPL14A	0	1	0	1	0	1	...
SST2	0	1	0	1	0	1	...
DRS2	0	0	0	1	0	1	...
GIT1	0	1	0	1	0	0	...
CLN3	0	0	0	1	1	1	...
RPO21	1	0	0	0	1	1	...
BIT89	0	1	0	0	0	1	...
...	...	...	...	...	...	...	...

yes
yes
yes
yes
yes
yes
yes
no
no
no
no
no
no
...

An example of a binding site module

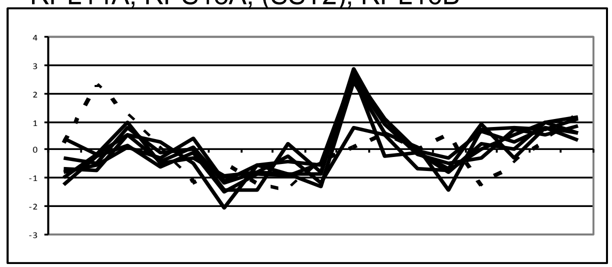
IF RAP1 **AND** SWI5 **AND** MCM1' **THEN**

The rule was (re-) discovered in five of the six expression data sets

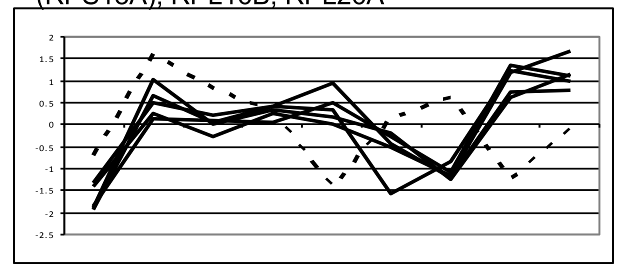
The central gene in the expression cluster is underlined

Genes with differing expression profiles are in parentheses

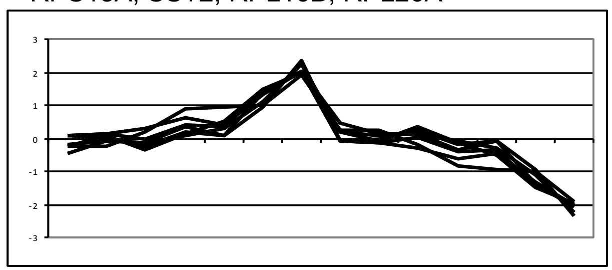
a) Cell cycle: RPL30, RPL18A, RPL26A, RPL14A, RPS18A, (SST2), RPL16B



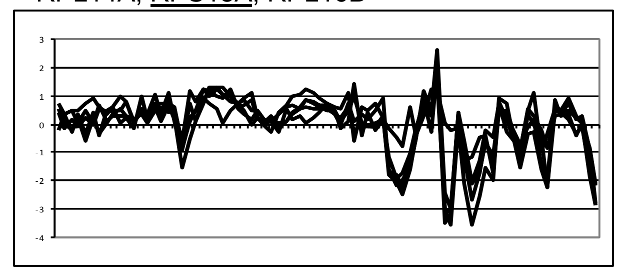
b) Sporulation: RPL30, RPL18A, RPL14A, (RPS18A), RPL16B, RPL26A



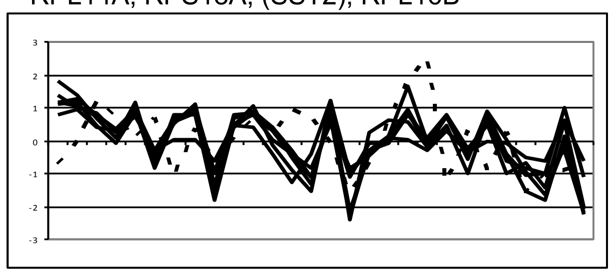
c) Diauxic shift: RPL30, RPL18A, RPL14A, RPS18A, SST2, RPL16B, RPL26A



d) Heat and cold shock: RPS24A, RPL26A, RPL14A, RPS18A, RPL16B



e) DNA-damaging agents: RPL30, RPL18A, RPL26A, RPL14A, RPS18A, (SST2), RPL16B



Biological significance

The transcription factor RAP1 targets two different sets of genes:

- genes that encode ribosomal proteins and that have an extremely high expression in rapidly growing yeast cells
- genes encoding several nonribosomal proteins

→ Combinatorial regulatory mechanism to separate these activities:

RAP1 specifically target ribosomal proteins in growing yeast cells by requiring the presence of cell cycle regulators MCM1 and SWI5.

Literature support:

- Gray and Fassler (1993): RAP1 forms a complex with MCM1
- Lydall et al. (1991): MCM1 and SWI5 are responsible for the cell-cycle-restricted transcription of SW15

Gasch, A.P. and Eisen, M.B. 2002. Exploring the conditional coregulation of yeast gene expression through fuzzy k-means clustering. Genome Biol. 3: 0059.0051-0059.0022.
Lydall, D., Ammerer, G., and Nasmyth, K. 1991. A new role for MCM1 in yeast: Cell cycle regulation of SW15 transcription. Genes & Dev. 5: 2405-2419

Biological significance cont: Gene Ontology

IF **RAP1 AND SWI5 AND MCM1'** THEN **<similar expression>**			
Gene symbol	**Biological process**	**Molecular function**	**Cellular component**
RPL16B	protein biosynthesis	RNA binding, structural constituent of ribosome	cytosolic ribosome (sensu Eukarya), large ribosomal subunit
RPL26A	protein biosynthesis	RNA binding, structural constituent of ribosome	cytosolic ribosome (sensu Eukarya), large ribosomal subunit
RPS18A	protein biosynthesis	structural constituent of ribosome	cytosolic ribosome (sensu Eukarya), eukaryotic 43S pre-initiation complex,eukaryotic 48S initiation complex, mall ribosomal subunit
RPL30	protein biosynthesis, rRNA processing, mRNA splicing, regulation of translation	structural constituent of ribosome	cytosolic ribosome (sensu Eukarya), cytoplasm, large ribosomal subunit
RPL18A	protein biosynthesis	structural constituent of ribosome	cytosolic ribosome (sensu Eukarya), large ribosomal subunit
RPL14A	protein biosynthesis	RNA binding, structural constituent of ribosome	cytosolic ribosome (sensu Eukarya), large ribosomal subunit
SST2	signal transduction, adaptation to pheromone during conjugation with cellular fusion	GTPase activator activity	plasma membrane
RPS24A	protein biosynthesis	structural constituent of ribosome	cytosolic ribosome (sensu Eukarya), eukaryotic 43S pre-initiation complex, eukaryotic 48S initiation complex, small ribosomal subunit
P-VALUE	2.35E-04 (*protein biosynthesis*)	2.36E-06 (*structural constituent of ribosome*)	5.66E-07 (*cytosolic ribosome*)

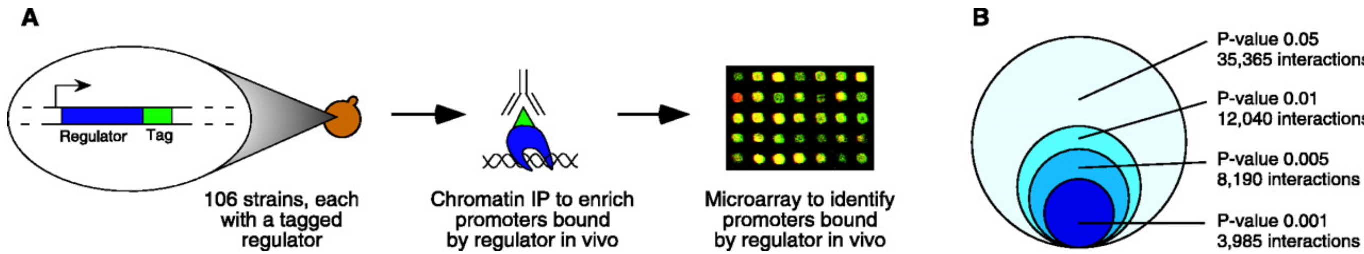
Statistical evaluation – Gene Ontology

- Fraction of significant rules (Bonferroni corrected P < 0.01)
- Comparison to: genes with similar expression, genes with common motifs and randomly sampled genes

Expression data	Gene Ontology evaluation (significant fractions P < 0.01)											
	Rule (P-values)			Randomized tests Similar expression | Common motifs | Random								
	Molecular function	Biological Process	Cellular component	Molecular Function			Biological Process			Cellular component		
Cell cycle	0.31 (0.000)	0.46 (0.000)	0.41 (0.000)	0.05	0.04	0.01	0.13	0.18	0.03	0.03	0.04	0.00
Sporulation	0.26 (0.000)	0.54 (0.000)	0.44 (0.000)	0.08	0.04	0.01	0.19	0.17	0.02	0.05	0.03	0.00
Diauxic Shift	0.30 (0.000)	0.43 (0.000)	0.44 (0.000)	0.04	0.05	0.02	0.11	0.17	0.03	0.02	0.03	0.00
Heat and cold shock	0.54 (0.000)	0.64 (0.006)	0.60 (0.000)	0.24	0.06	0.03	**0.46**	0.24	0.05	0.17	0.04	0.01
Pheromone	0.51 (0.000)	0.67 (0.000)	0.60 (0.000)	0.10	0.05	0.01	0.25	0.16	0.02	0.08	0.03	0.00
DNA-damaging agents	0.39 (0.000)	0.64 (0.000)	0.61 (0.000)	0.09	0.05	0.01	0.19	0.17	0.03	0.07	0.04	0.00

Genome-wide location analysis

Lee et al. (2002) experimentally measured the probability of 106 yeast transcription factors binding to gene promoters



Genes	Transcription factors												
	ABF1	ACE2	ADR1	ARG80	ARG81	ARO80	ASH1	AZF1	BAS1	CAD1	CBF1	HIR1	...
YAL001C	0.580	0.730	0.490	0.240	0.050	0.300	0.600	0.250	0.061	1.000	1.000	***0.005***	...
YAL002W	1.000	0.760	0.560	0.150	0.079	0.160	1.000	0.012	0.620	1.000	1.000	0.220	...
YAL015C	1.000	0.210	0.130	0.250	0.190	0.340	1.000	0.410	0.410	1.000	1.000	0.400	...
YAL016W	***0.002***	0.920	0.170	0.480	0.340	0.320	0.690	0.150	0.580	0.310	0.540	0.630	...
YAL017W	0.970	0.048	***0.003***	0.980	0.840	0.970	0.940	0.610	0.870	0.210	0.340	0.440	...
YAL018C	0.970	0.048	***0.003***	0.980	0.840	0.970	0.940	0.610	0.870	0.210	0.340	0.440	...
YAL019W	0.600	0.360	0.011	0.900	0.830	0.700	0.170	0.940	0.650	0.380	0.051	0.590	...
YAL020C	0.600	0.360	0.011	0.900	0.830	0.700	0.170	0.940	0.650	0.380	0.051	0.590	...
YAL021C	0.410	0.340	0.160	0.400	0.720	0.410	0.590	0.960	0.840	0.630	0.017	0.170	...
YAL022C	0.810	0.190	***0.001***	0.970	0.870	0.350	0.530	0.790	0.930	0.500	0.085	0.047	...
YAL023C	***0.000***	0.930	0.380	0.430	0.770	0.280	0.330	0.840	0.840	0.400	0.041	0.730	...
...	...	...	...	...	...	...	...	...	...	...	...	...	...

Lee et al. 2002. Transcriptional regulatory networks in Saccharomyces cerevisiae. *Science* **298:** 799-804.

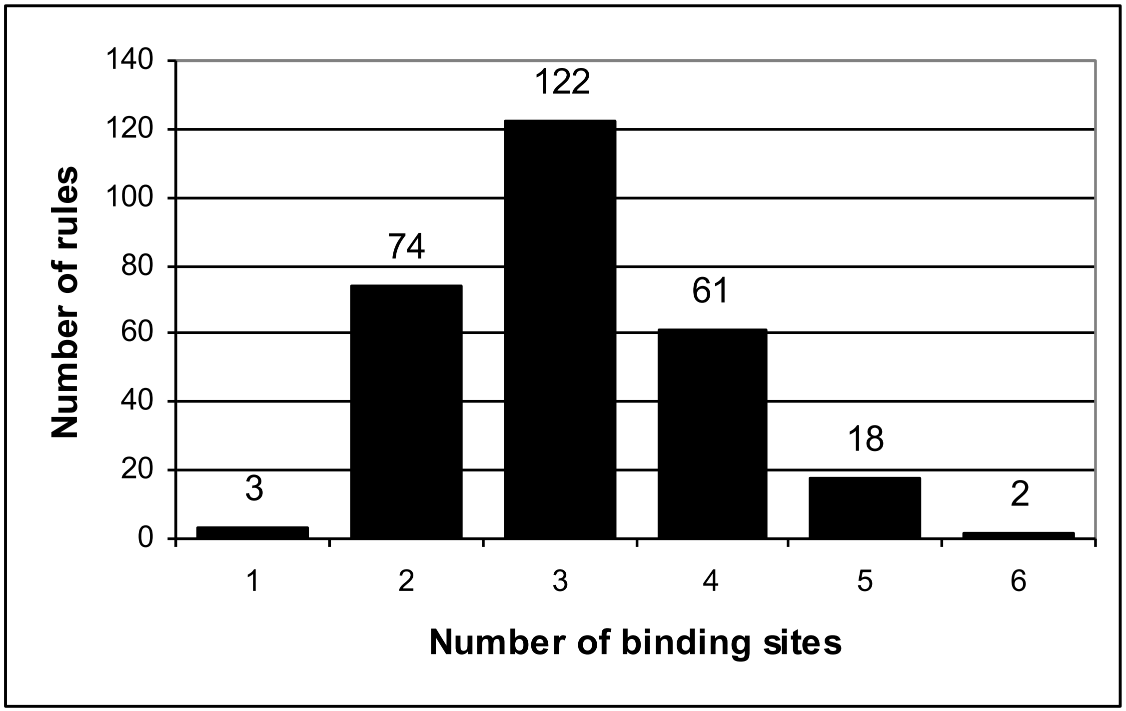
Statistical evaluation – TF binding

- Fraction of significant rules (Bonferroni corrected $P < 0.01$)
- Comparison to: genes with similar expression, genes with common motifs and randomly sampled genes

Expression data	Expression similarity thresholds	No. rules unique/all	Binding data evaluation (significant fractions $P < 0.01$)			
			Rules (P-value)	Randomized tests		
				Similar expression	Common motifs	Random
Cell cycle	0.250	39/109	0.54 (0.000)	0.11	0.17	0.02
Sporulation	0.250	45/81	0.13 (0.708)	0.09	**0.18**	0.02
Diauxic shift	0.200	150/428	0.29 (0.000)	0.06	0.18	0.02
Heat and cold shock	0.125	52/123	0.52 (0.000)	0.18	0.18	0.02
Pheromone	0.150	53/91	0.39 (0.001)	0.14	**0.17**	0.02
DNA-damaging agents	0.200	59/116	0.35 (0.000)	0.10	0.17	0.02

Module complexity

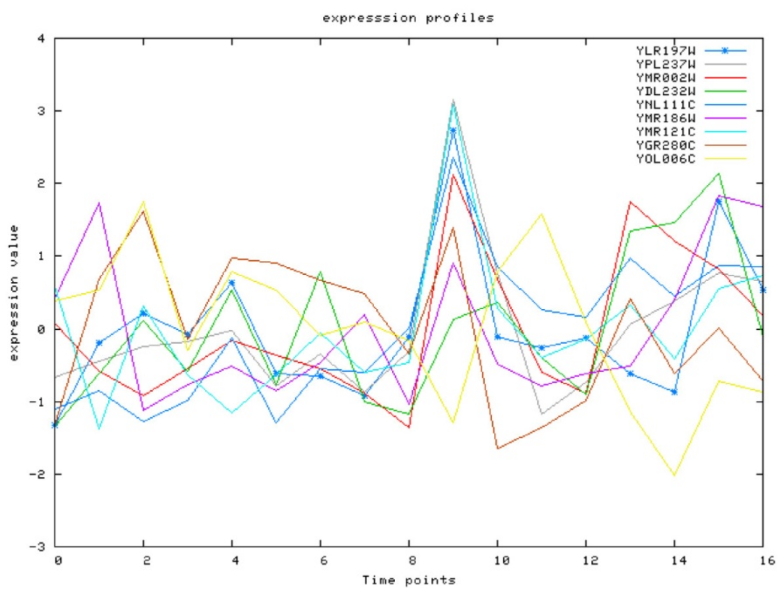
The distribution of the number of binding sites in the discovered modules



Restricting expression similarity to time-windows

IF REB1 **AND** SWI5 **AND** SCB

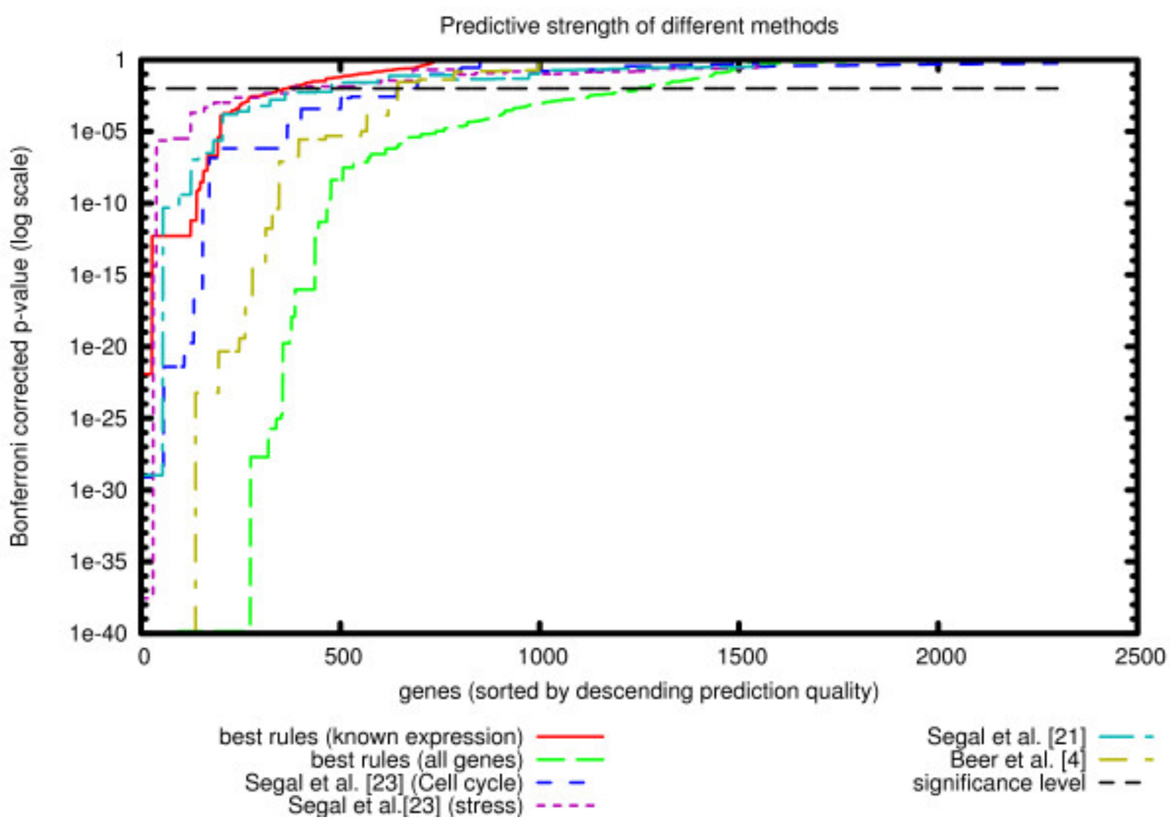
[9–10]~0.9



- Not identified using global expression similarity
- P-value for the ChIP-Chip data: $4 \cdot 10^{-6}$
- Corresponds to the M1/G phase boundary which is the active time of the SWI5 factor

B. Wilczynski, T. R. Hvidsten, A. Kryshtafovych, J. Tiuryn, J. Komorowski, K. Fidelis. Using Local Gene Expression Similarities to Discover Regulatory Binding Site Modules, BMC Bioinformatics 7: 505, 2006.

Explanatory power



- ChIP-Chip data
- Cell cycle expression data
- Number of genes explained by significant rules
- Comparison to other studies

Model-based detection of periodic expression: using biological knowledge

C. R. Andersson*, T. R. Hvidsten*, A. Isaksson, M. G. Gustafsson, J. Komorowski. Revealing cell cycle control by combining model-based detection of periodic expression with cis-regulatory descriptors, Accepted to BMC Systems Biology, 1: 45, 2007.

* Contributed equally

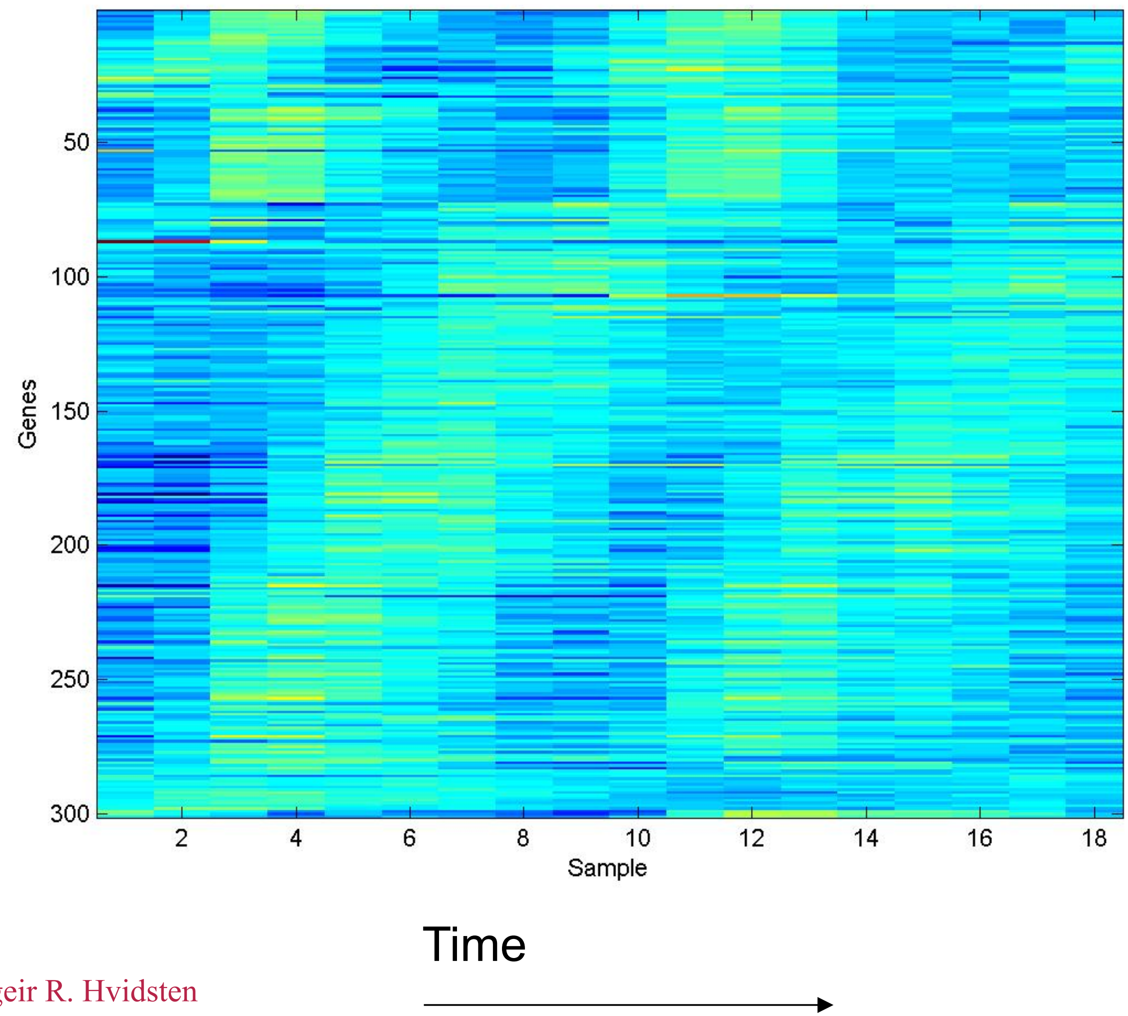
Synchronization

- To recover mRNA-levels, cultures must be synchronized.
- Synchronization halts cells at a particular point in the cell cycle.
- Typically the mating system or temperature sensitive mutants are used (cdc28, cdc15).

- Spellman (1998). *S cerevisae* under various synchronizations (alpha-factor, ts cdc15, cdc28 and elutriation)

Spellman PT, Sherlock G, Zhang MQ, Iyer KA, Eisen MB, Brown PO, Botstein D, Futcher B, Comprehensive identication of cell cycle-regulated genes of the yeast Saccharomyces cerevisiae by microarray hybridization . Mol Biol Cell 1998, 9:3273-97.

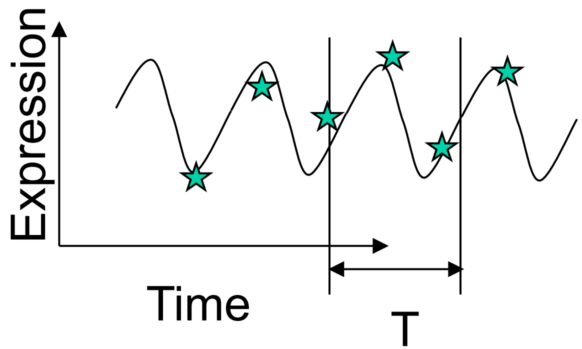
Synchronization cont.



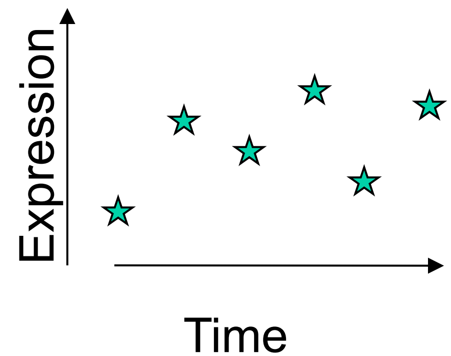
- Synchronization reveals periodic expression
- Periodic expression is related to the cell cycle machinery

Detecting periodically expressed genes

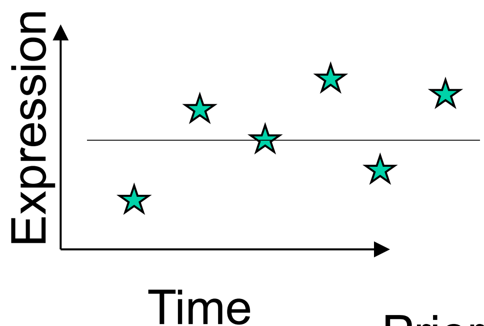
Model 1



Signal



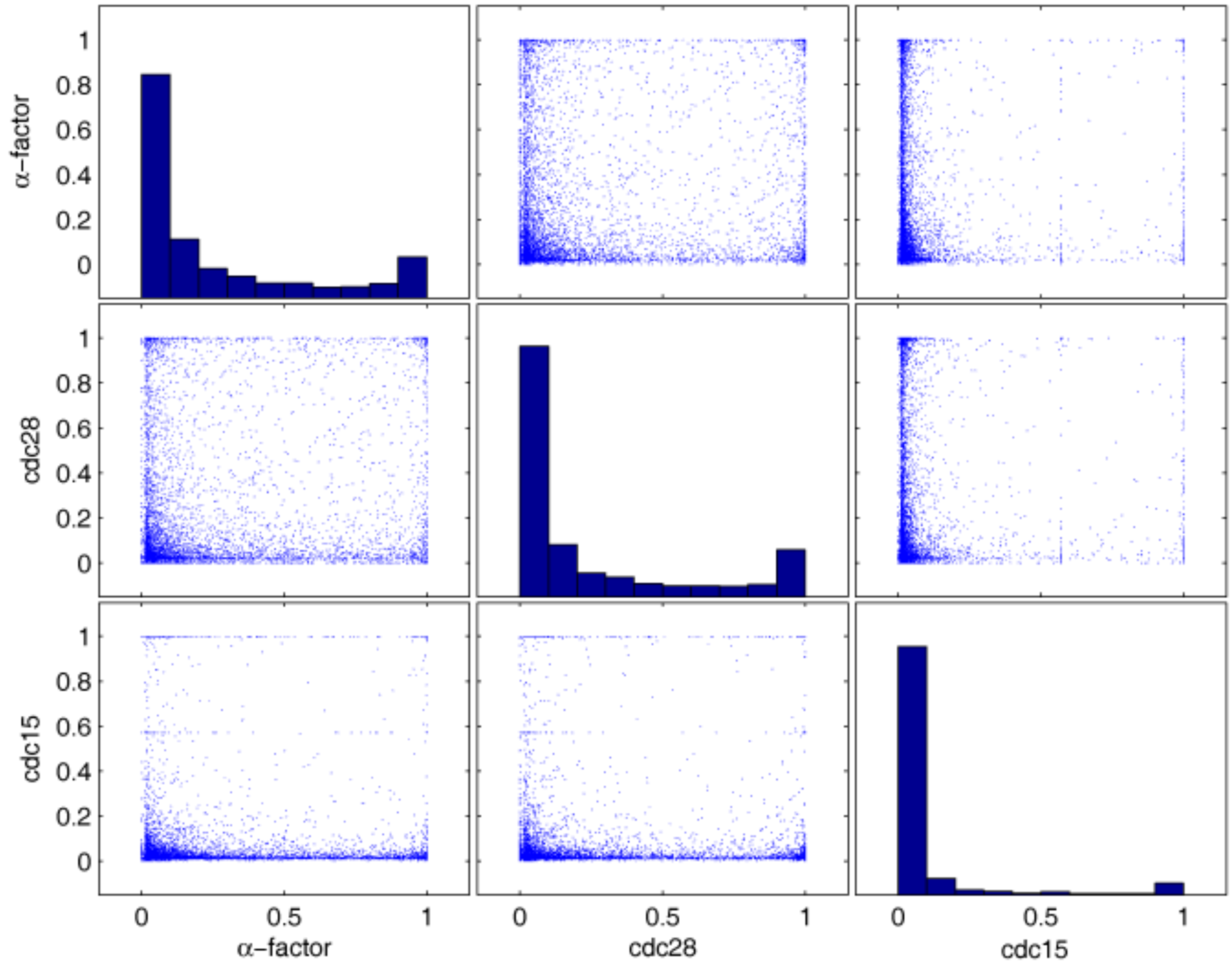
Model 0



Which model is most similar to the signal?
→
Probability (Periodical expression)

Prior knowledge:
Period time of the cell cycle in Yeast

Conditional periodicity



Model-based analysis of gene regulation

- Rather than relying on clustering, apply a model to describe the gene expression
  - Prior knowledge:
    - Period time of the cell cycle
    - Periodic expression is related to the cell cycle machinery
- Advantage:
  - Interpretable hypotheses
  - More specific hypotheses

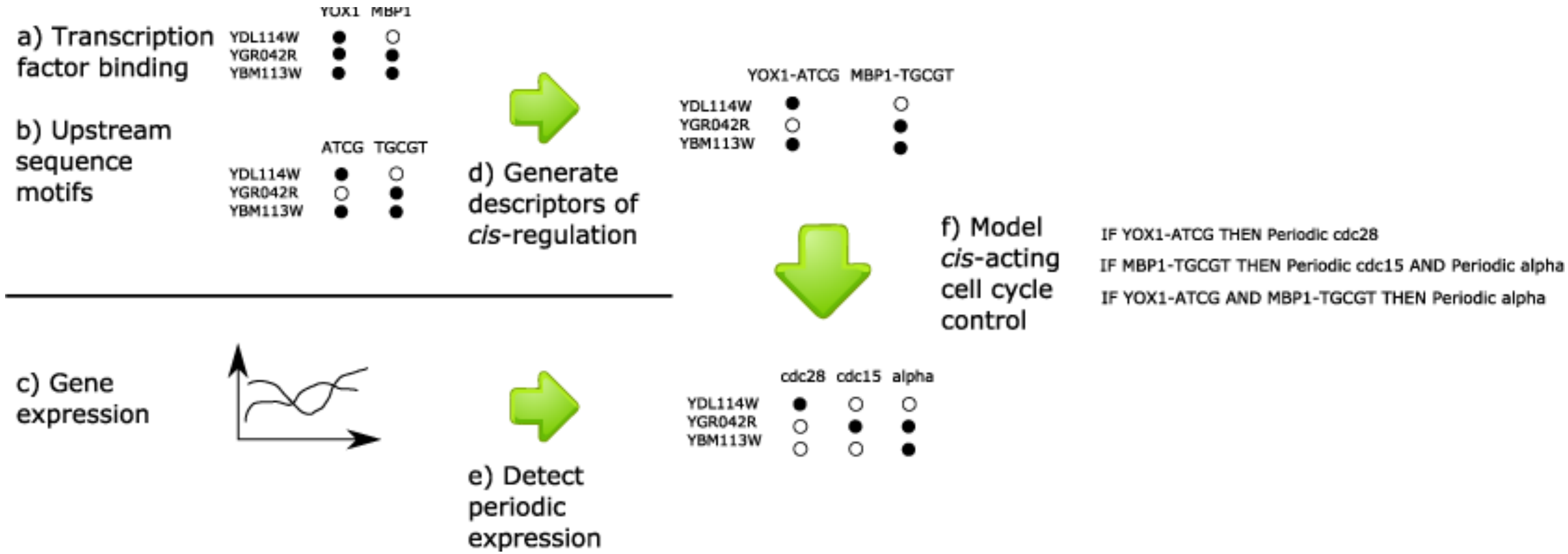
Periodic classes

Classication with rejection:

- P(Periodic) > 0.90: Periodic
- P(Periodic) < 0.10: Not periodic

Class	α-factor	Cdc28	cdc15	No. genes
000	Not periodic	Not periodic	Not periodic	1173
001	Not periodic	Not periodic	Periodic	55
010	Not periodic	Periodic	Not periodic	140
100	Periodic	Not periodic	Not periodic	4
011	Not periodic	Periodic	Periodic	127
101	Periodic	Not periodic	Periodic	11
110	Periodic	Periodic	Not periodic	115
111	Periodic	Periodic	Periodic	19
Sum				1644

Method



a) Transcription factor binding
YOX1 MBP1
YDL114W
YGR042R
YBM113W
b) Upstream sequence motifs
ATCG TGCGT
YDL114W
YGR042R
YBM113W
c) Gene expression
d) Generate descriptors of *cis*-regulation
YOX1-ATCG MBP1-TGCGT
YDL114W
YGR042R
YBM113W
e) Detect periodic expression
cdc28 cdc15 alpha
YDL114W
YGR042R
YBM113W
f) Model *cis*-acting cell cycle control
IF YOX1-ATCG THEN Periodic cdc28
IF MBP1-TGCGT THEN Periodic cdc15 AND Periodic alpha
IF YOX1-ATCG AND MBP1-TGCGT THEN Periodic alpha

Cis-regulatory descriptors

Transcription factor binding and sequence motifs co-occurring in promoter regions

- Overrepresented $p < 0.05$
- Three best pairs for each factor
- Three best pairs for each motif

Known associations

TF	P-value	Motif
YOX1	2.25e-06	MCM1
	3.83e-05	ECB
	0.0034	m_organization_of_cell wall_orfnum2SD_n6
UME6	5.70e-114	m_meiosis_orfnum2SD_n3
	1.75e-70	Ume6(URS1)
	1.74e-17	m_glyoxylate_cycle_orfnum2SD_n11
ABF1	6.5e-214	ABF1
	2.98e-09	Ume6(URS1)
	7.8e-09	RPN4

Cis-regulatory descriptors cont

- 1459 *cis*-regulatory descriptors in total
- Remember: 1644 genes

Motif	**P-value**	**TF**
SCB	3.67e-08	*AZF1*
	1.02e-05	*UME6*
	3.31e-05	*SWI4*
SFF	1.14e-10	*FKH2*
	9.26e-10	*FKH1*
	1.92e-05	*HIR1*

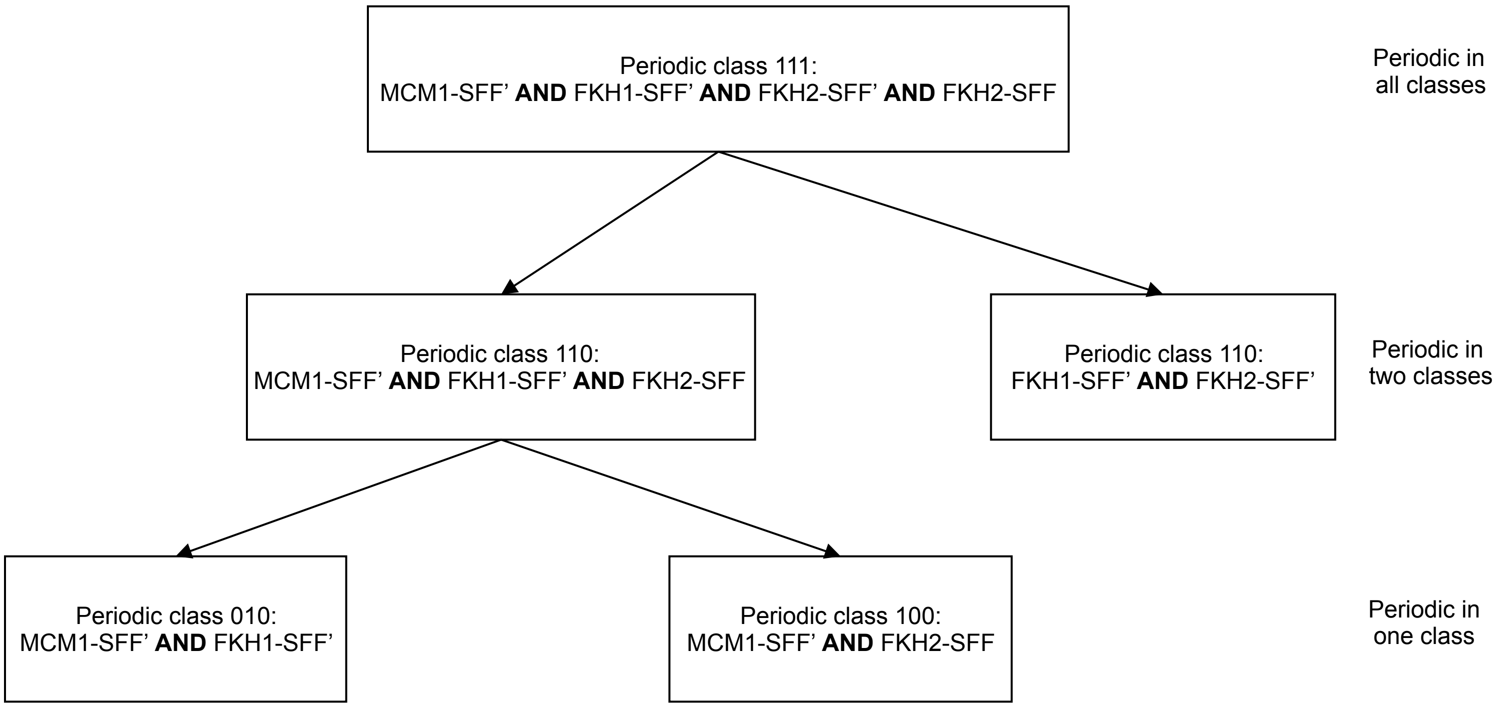
Results

- Pairs of cis-regulatory descriptors are better at explaining the periodic classes than single descriptors.
- Pairs are specific to different periodic classes: indicate that the synchronization methods induce different perturbations that initially activate different regulatory mechanisms visible in the two first periods of the cell cycle.

Class	Observed/Expected (P-value) - Single	Observed/Expected (P-value) - Pairs		No. genes
		All genes	Within classes	
001	1.1 (0.35)	7.3 (5e-10)	0.86 (0.71)	55
010	0.77 (0.9)	0.95 (0.91)	0.31 (1.0)	140
100	0.85 (0.84)	0.36 (1.0)	0.0 (1.0)	127
011	9.1 (1.3e-11)	19.3 (<1e-20)	18.18 (1.02e-12)	4
101	4.42 (1.55e-9)	19.1 (2.5e-12)	12.9 (1.3e-14)	11
110	2.2 (1.3e-10)	1.6 (1.44e-10)	1.81 (0.0013)	115
111	3.7 (4.79e-13)	17.1 (<1e-20)	6.7 (3.8e-14)	19

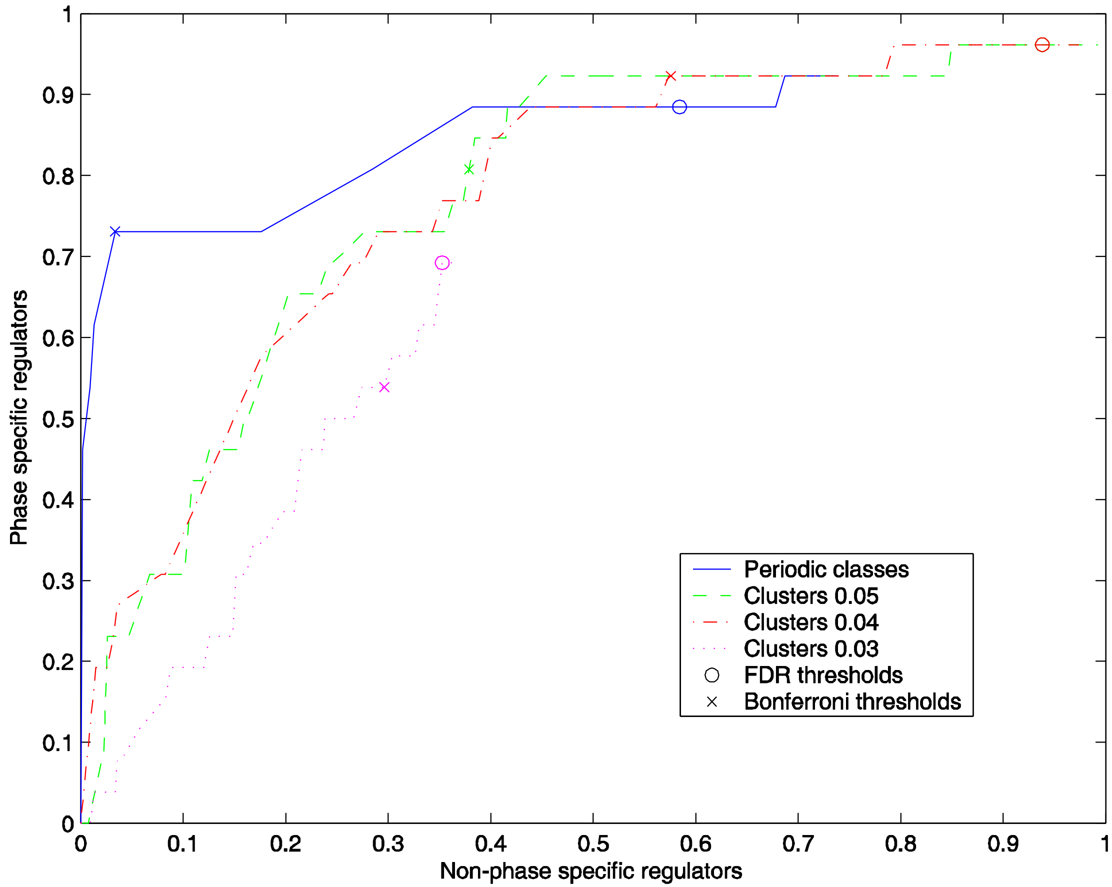
Results cont.

- Combinations of regulatory descriptors suggests that the periodic classes are regulated in an additive fashion.



Results cont.

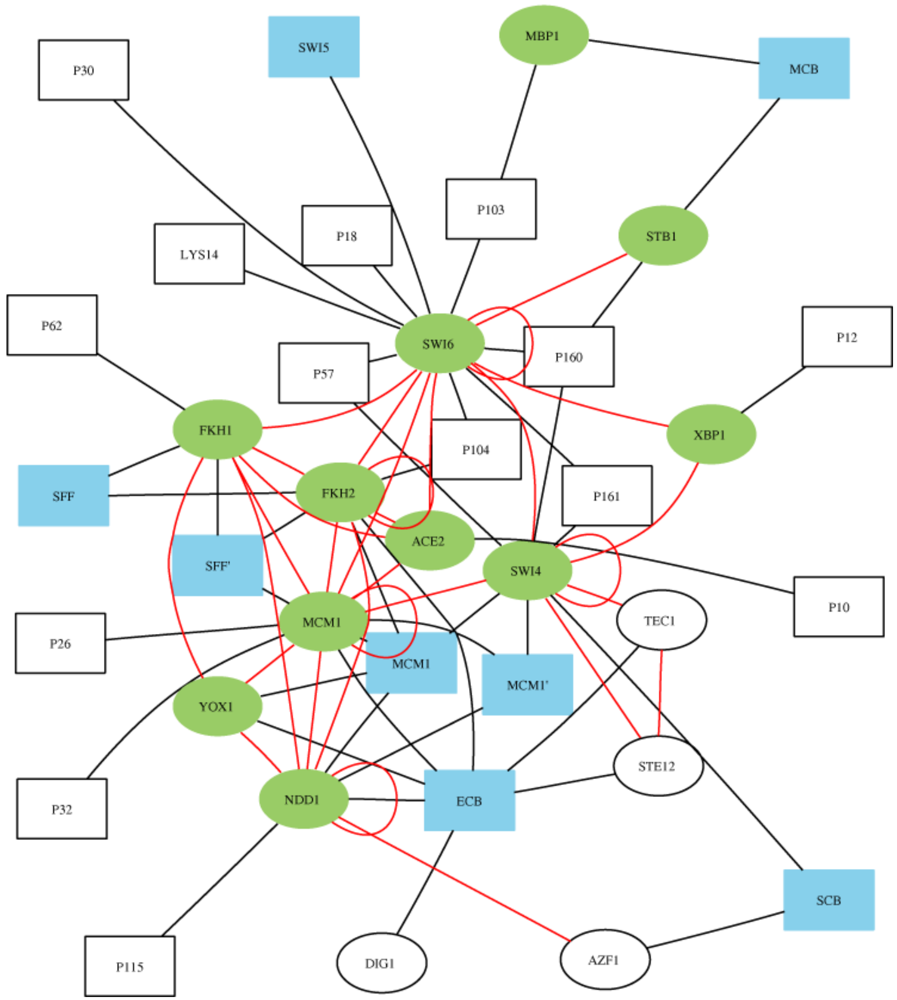
- Replacing clustering-based classification of dynamic gene expression patterns with model-based classification is advantageous for discovering the mechanisms underlying cellular control processes.



Explanatory power

- The point (0.034, 0.73) on the curve is associated with the 145 rules with p-value lower than 0.000195.
- These rules include 19 of the 26 known phase specific regulators (73%) and 18 other regulators 3.4%.
- Furthermore, they describe 24% of the genes in the periodic classes.

Examples of ”interactions”



Ellipses/rectangles: transcription factors/ sequence motifs

Green/Blue: Cell cycle related transcription factors/sequence motifs

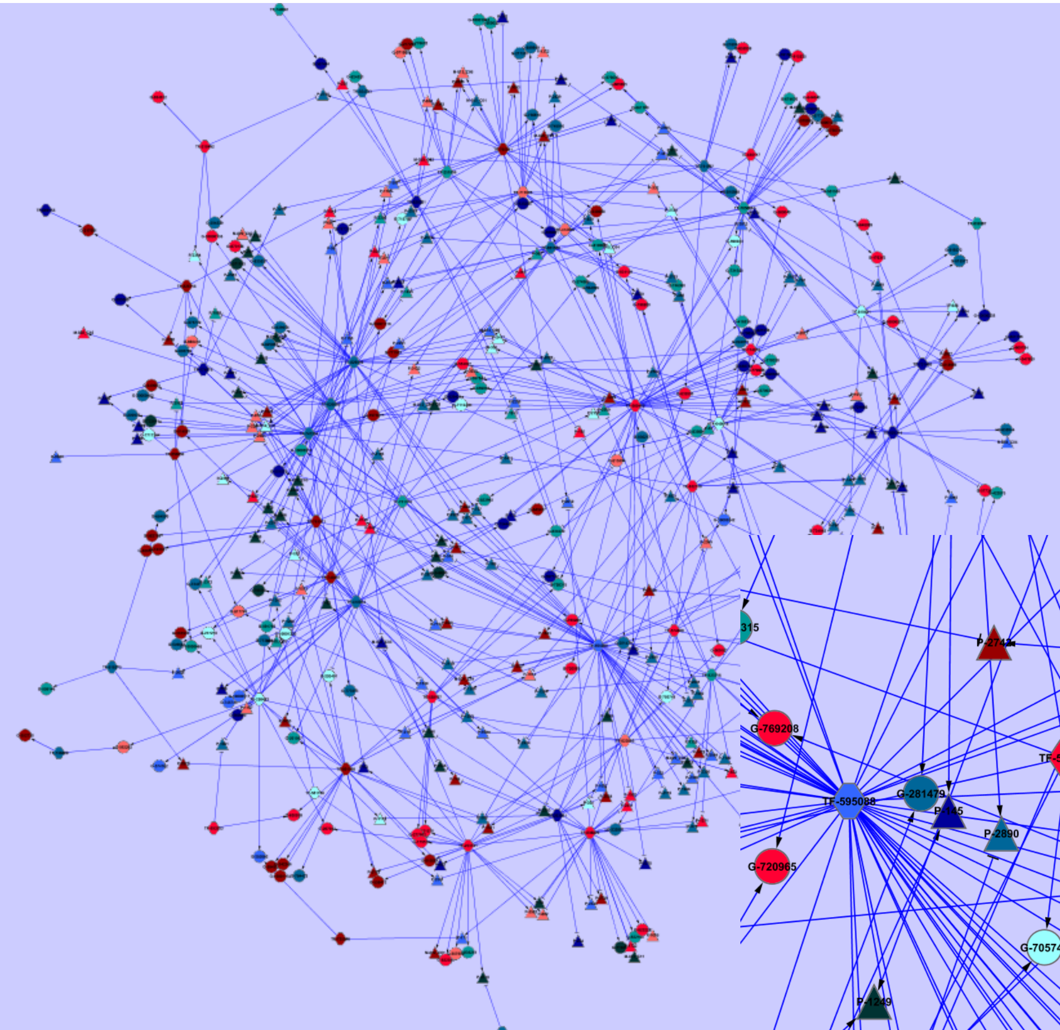
Reed: ”interactions” between transcription factors

Future challenges:
towards regulatory networks

Modules combined into networks

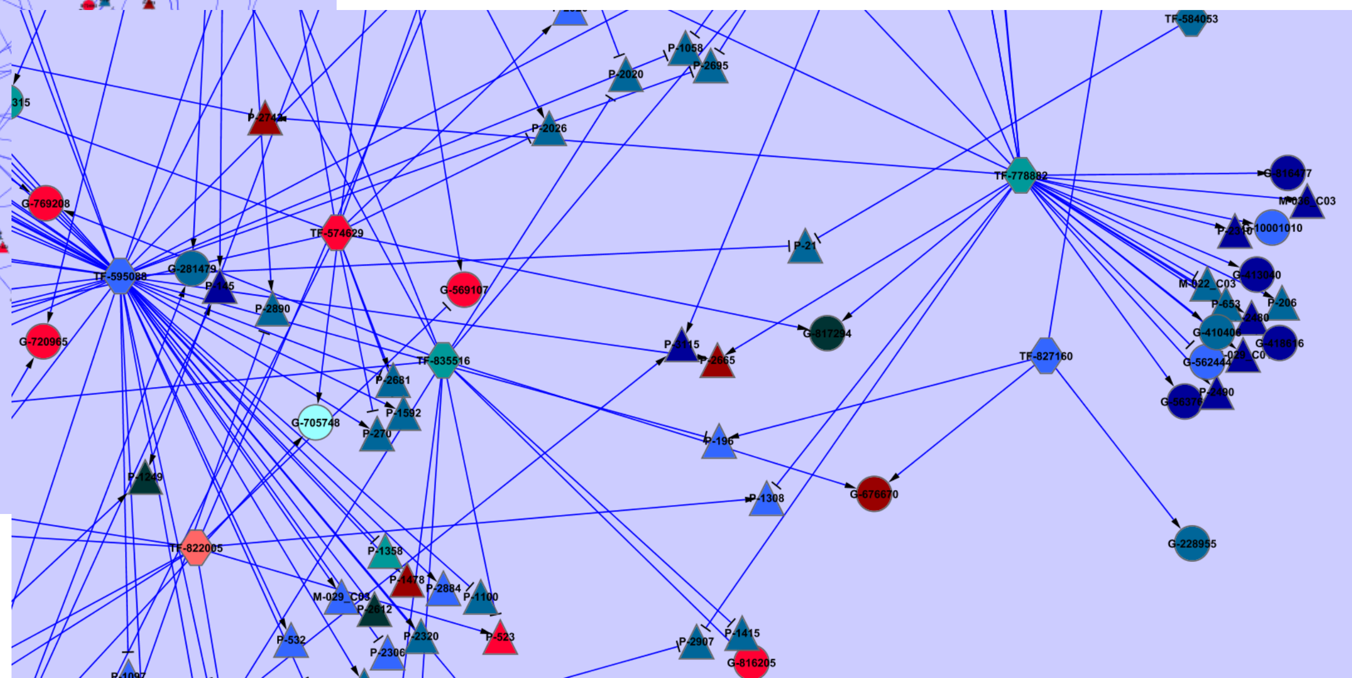
- Modules reduce the number of nodes in the network
- The regulatory mechanisms predicted for each module can further restict the potential regulators
- The expression of predicted transcription factors can be used to link modules together (e.g. Using Bayesian networks)

Preliminary results: trees (Populus/aspen)



Three platforms to explain regulation in trees:

1. Transcriptomics: **G**enes and **T**ranscription **F**actors
2. Metabolomics: **M**etabolites
3. Proteomics: **P**roteins



Machine learning issues

- Representation
  - – Expression trends over time intervals
  - – Expression similarity as decision classes
- Data integration
  - Expression data – protein sequence features
  - Sequence motifs – expression data
  - Sequence motifs – binding data
- Knowledge integration
  - Gene Ontology
  - Periodic expression targets the cell cycle macinery

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